HCI STAGE 1 — SECTION CINE-DECK

Dataset: ember_JPN_Japan_generation_TWh.csv

Carriers: D = 8 (Bioenergy, Coal, Gas, Hydro, Nuclear, Other Fossil, Solar, Wind)

Observations: T = 26 (2000–2025)

Bearing pairs: 28

Plates: 26 section triads (1 per year)

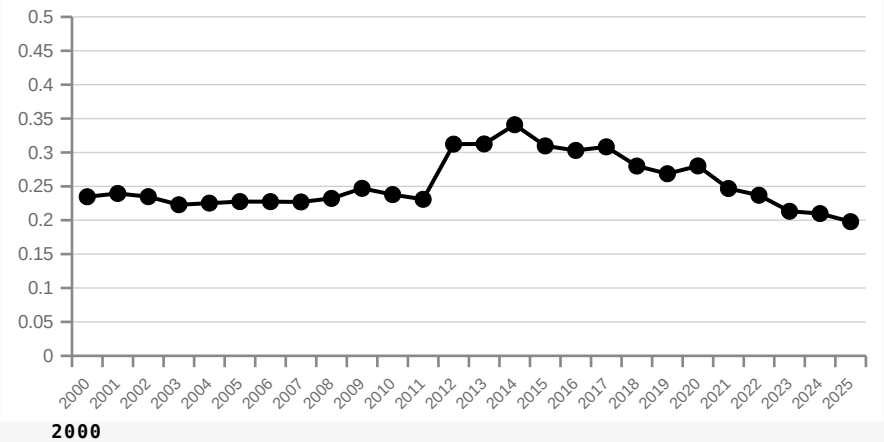
Engine: Compositional Navigation Tensor (CNT)

The instrument reads. The expert decides. The loop stays open.

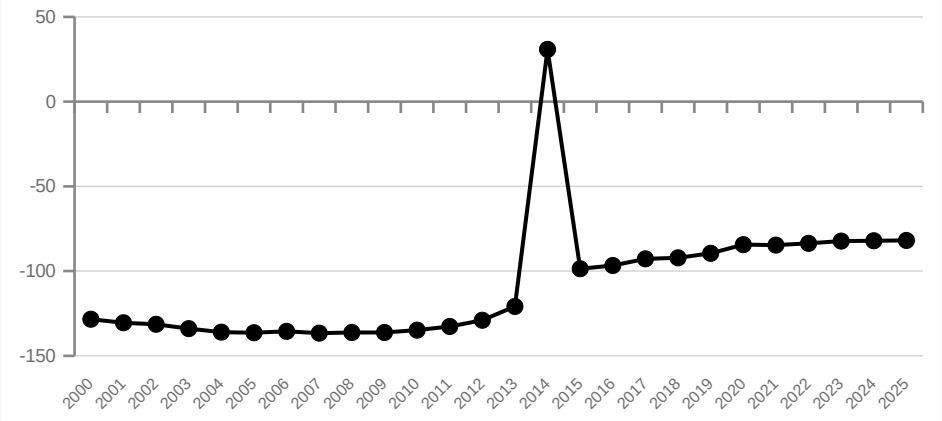
SECTION PLATE t=0 | Year 2000

Hs = 0.2347 | Hs-2 | omega = 0.00 deg

YZ: Hs vs Time



XZ: Bearing (Sola-Wind) vs Time



XY PLAN VIEW: All Bearings (deg)

Solar-Wind	-128.4	Nuclear-Wind	-65.1
Bioenergy-Hydro	+111.1	Hydro-Nuclear	+64.7
Bioenergy-OFos	+103.2	Coal-Solar	-62.7
Bioenergy-Coal	+101.7	Gas-Solar	-61.8
Bioenergy-Gas	+101.3	Nuclear-Solar	-59.7
Bioenergy-Nuclear	+100.4	Hydro-OFos	+58.7
Bioenergy-Solar	-96.1	Coal-Nuclear	+48.5
Bioenergy-Wind	-94.8	Gas-Nuclear	+47.5
Hydro-Wind	-77.6	Coal-Gas	+46.0
Hydro-Solar	-74.5	Coal-OFos	+41.4
OFos-Wind	-70.2	Gas-OFos	+40.4
Coal-Wind	-67.7	Nuclear-OFos	+37.9
Gas-Wind	-67.0	Coal-Hydro	+28.2
OFos-Solar	-65.5	Gas-Hydro	+27.3

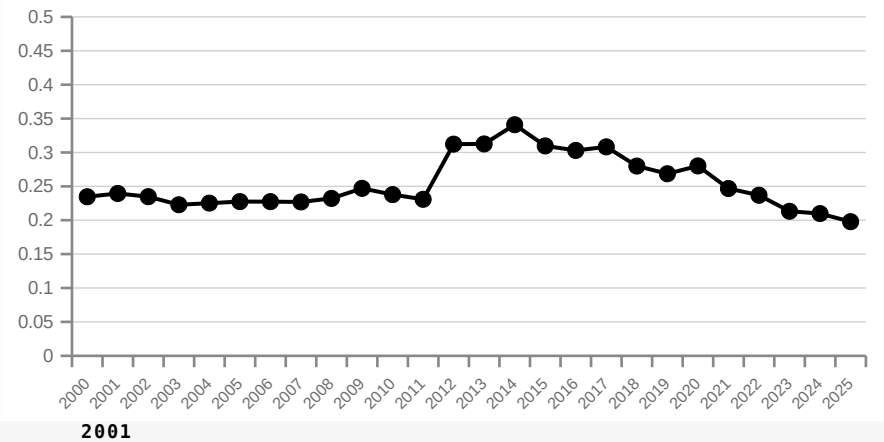
METRIC LEDGER

```
Hs     = 0.234683    Ring = Hs-2
E_metric = 70.567287    kappa = 2901.09
omega    = 0.0000 deg
d_A     = 0.000000
Helm     = --- delta = +0.000000

CLR:
 Bio -0.462  Coal +2.232  Gas +2.313  Hyd +1.195
 Nuc +2.524  OFos +1.967  Sol -4.320  Wind -5.449

Steering (1/x_j):
 Bio   68.3 Coal   4.6 Gas    4.3 Hyd   13.0
 Nuc    3.4 OFos   6.0 Sol   3e+3 Wind  1e+4
```

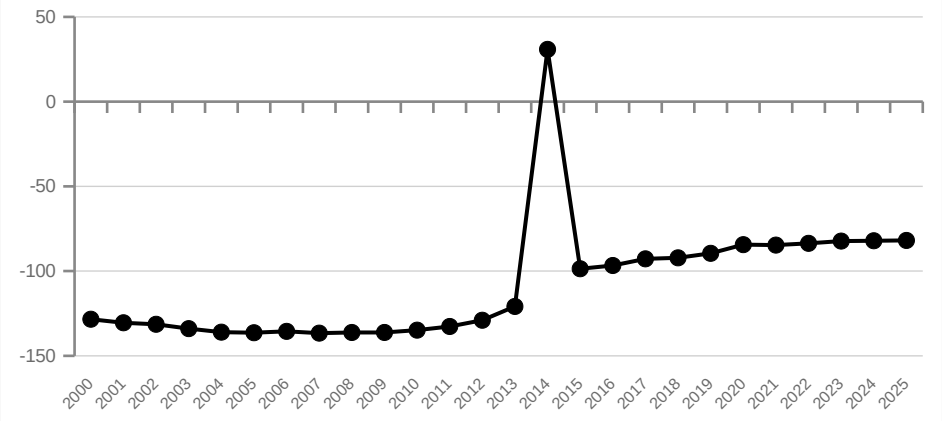
SECTION PLATE t=1 | Year 2001

Hs = 0.2396 | Hs-2 | omega = 2.82 deg

YZ: Hs vs Time



XZ: Bearing (Sola-Wind) vs Time



XY PLAN VIEW: All Bearings (deg)

Solar-Wind	-130.5	Gas-Wind	-65.5
Bioenergy-Hydro	+120.2	Nuclear-Wind	-63.3
Bioenergy-OFos	+109.8	Coal-Solar	-62.0
Bioenergy-Coal	+105.5	Gas-Solar	-61.9
Bioenergy-Gas	+105.4	Nuclear-Solar	-59.5
Bioenergy-Nuclear	+104.0	Hydro-OFos	+58.3
Bioenergy-Solar	-98.4	Coal-Nuclear	+47.9
Bioenergy-Wind	-97.2	Gas-Nuclear	+47.8
Hydro-Wind	-77.8	Coal-Gas	+45.1
Hydro-Solar	-75.8	Coal-OFos	+37.5
OFos-Wind	-70.7	Gas-OFos	+37.4
OFos-Solar	-67.8	Nuclear-OFos	+34.8
Hydro-Nuclear	+66.8	Coal-Hydro	+25.4
Coal-Wind	-65.6	Gas-Hydro	+25.3

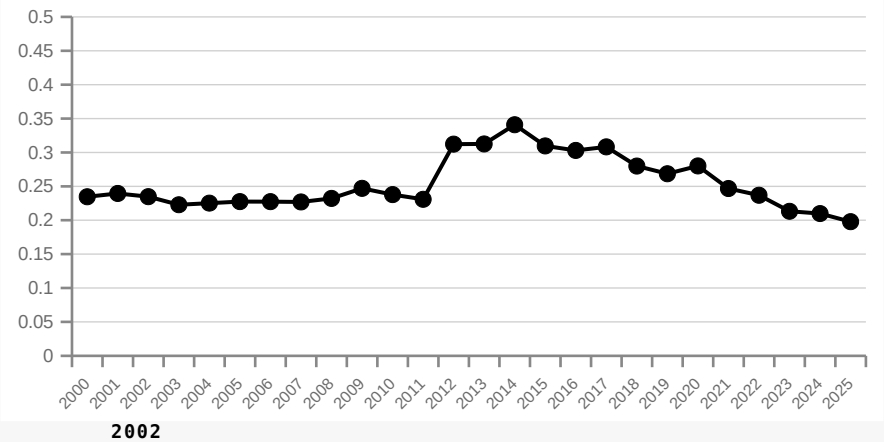
METRIC LEDGER

```
Hs     = 0.239561    Ring = Hs-2
E_metric = 58.520782    kappa = 1282.16
omega    = 2.8189 deg
d_A     = 0.847846
Helm     = Wind delta = +0.689167

CLR:
 Bio -0.599  Coal +2.164  Gas +2.174  Hyd +1.028
 Nuc +2.397  OFos +1.663  Sol -4.066  Wind -4.760

Steering (1/x_j):
 Bio   67.6 Coal   4.3 Gas   4.2 Hyd   13.3
 Nuc    3.4 OFos   7.0 Sol  2e+3 Wind  4e+3
```

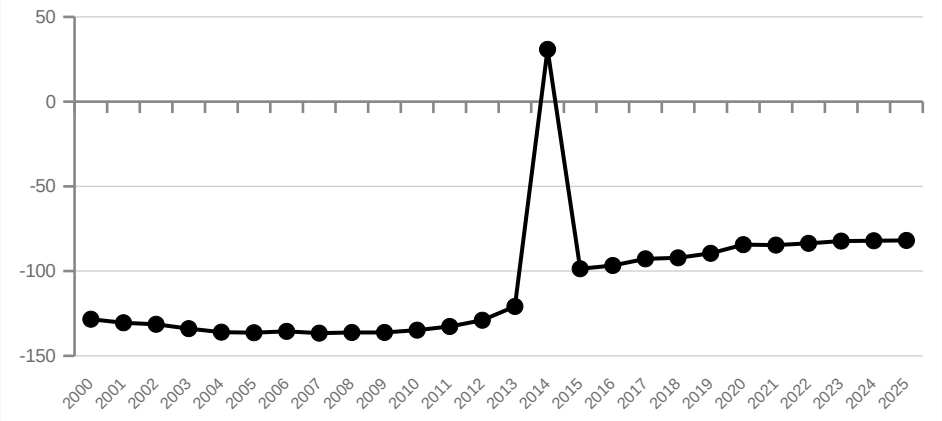
SECTION PLATE t=2 | Year 2002

Hs = 0.2348 | Hs-2 | omega = 1.52 deg

YZ: Hs vs Time



XZ: Bearing (Sola-Wind) vs Time



XY PLAN VIEW: All Bearings (deg)

Solar-Wind	-131.4	Coal-Wind	-64.6
Bioenergy-Hydro	+126.7	Nuclear-Wind	-62.8
Bioenergy-OFos	+111.9	Gas-Solar	-62.2
Bioenergy-Gas	+108.1	Coal-Solar	-61.7
Bioenergy-Coal	+107.8	Hydro-OFos	+61.7
Bioenergy-Nuclear	+106.5	Nuclear-Solar	-59.8
Bioenergy-Solar	-99.8	Gas-Nuclear	+47.9
Bioenergy-Wind	-98.6	Coal-Nuclear	+47.3
Hydro-Wind	-78.5	Coal-Gas	+44.4
Hydro-Solar	-77.0	Gas-OFos	+39.1
OFos-Wind	-69.3	Coal-OFos	+38.5
Hydro-Nuclear	+68.4	Nuclear-OFos	+36.3
OFos-Solar	-66.8	Gas-Hydro	+23.7
Gas-Wind	-65.1	Coal-Hydro	+23.2

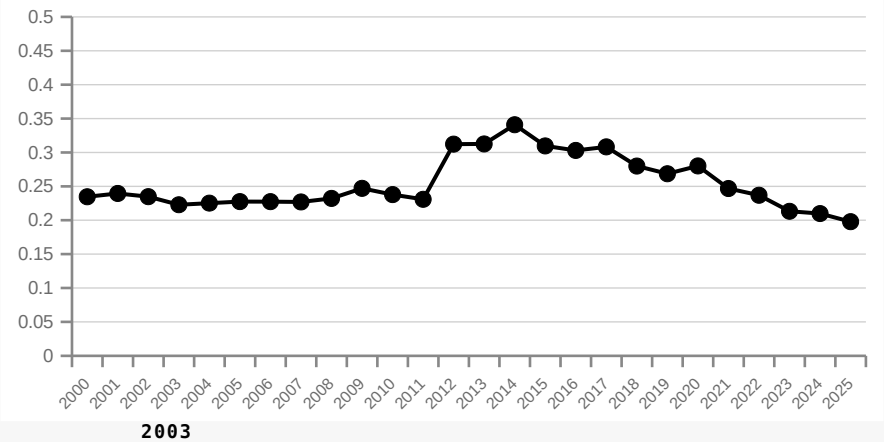
METRIC LEDGER

```
Hs     = 0.234830   Ring = Hs-2
E_metric = 51.755351   kappa = 766.49
omega   = 1.5248 deg
d_A     = 0.496689
Helm    = Wind delta = +0.371738

CLR:
 Bio -0.667  Coal +2.081  Gas +2.036  Hyd +0.893
 Nuc +2.254  OFos +1.657  Sol -3.867  Wind -4.388

Steering (1/x_j):
 Bio   65.1 Coal   4.2 Gas   4.4 Hyd   13.7
 Nuc    3.5 OFos   6.4 Sol  2e+3 Wind  3e+3
```

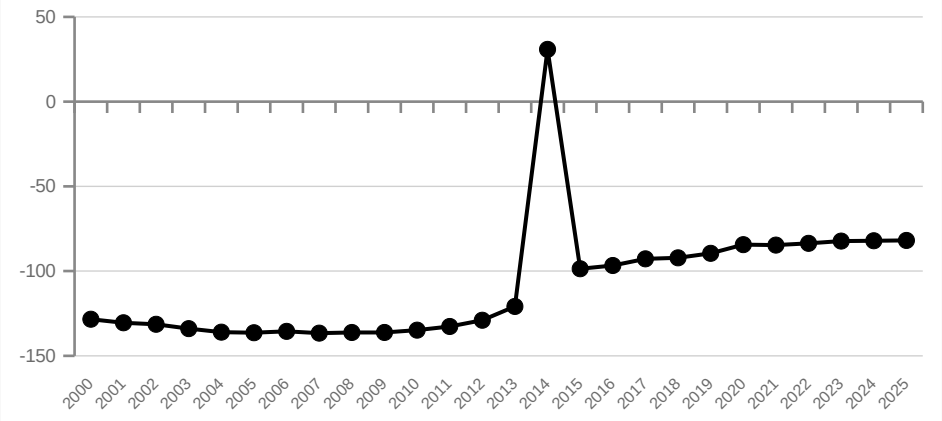
SECTION PLATE t=3 | Year 2003

Hs = 0.2228 | Hs-2 | omega = 3.77 deg

YZ: Hs vs Time



XZ: Bearing (Sola-Wind) vs Time



XY PLAN VIEW: All Bearings (deg)

Solar-Wind	-134.0	Hydro-Nuclear	+63.7
Bioenergy-Hydro	+130.5	Gas-Wind	-62.8
Bioenergy-OFos	+115.6	Coal-Wind	-62.0
Bioenergy-Nuclear	+112.8	Gas-Solar	-61.9
Bioenergy-Gas	+111.0	Coal-Solar	-61.1
Bioenergy-Coal	+110.4	Hydro-OFos	+60.7
Bioenergy-Solar	-101.6	Coal-Gas	+44.0
Bioenergy-Wind	-101.2	Gas-Nuclear	+42.4
Hydro-Wind	-76.9	Coal-Nuclear	+41.4
Hydro-Solar	-76.5	Nuclear-OFos	+41.3
OFos-Wind	-67.5	Gas-OFos	+38.8
OFos-Solar	-66.8	Coal-OFos	+37.8
Nuclear-Wind	-64.8	Gas-Hydro	+24.2
Nuclear-Solar	-64.0	Coal-Hydro	+23.5

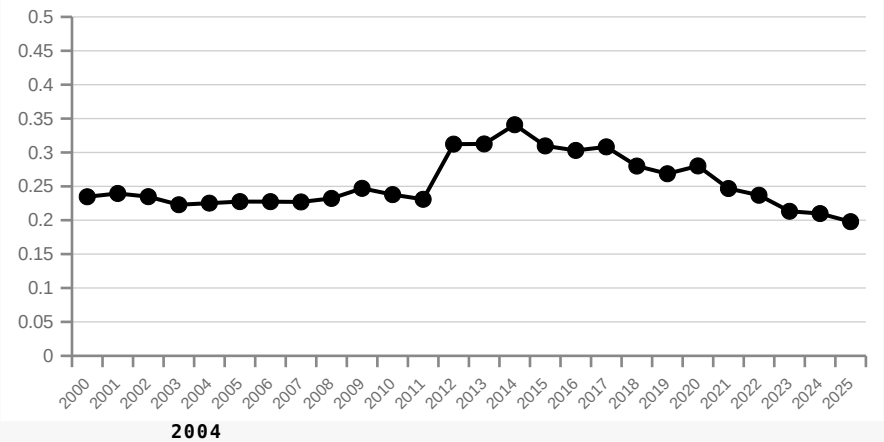
METRIC LEDGER

```
Hs     = 0.222849   Ring = Hs-2
E_metric = 43.390111   kappa = 351.88
omega   = 3.7738 deg
d_A     = 0.757601
Helm    = Wind delta = +0.561548

CLR:
 Bio -0.756  Coal +2.037  Gas +1.970  Hyd +0.887
 Nuc +1.798  OFos +1.582  Sol -3.691  Wind -3.826

Steering (1/x_j):
 Bio   61.1 Coal   3.7 Gas    4.0 Hyd   11.8
 Nuc    4.7 OFos   5.9 Sol  1e+3 Wind  1e+3
```

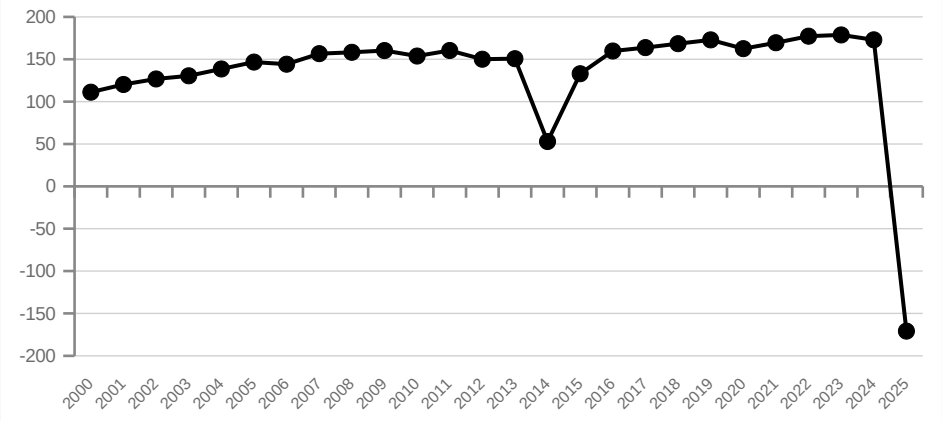
SECTION PLATE t=4 | Year 2004

Hs = 0.2253 | Hs-2 | omega = 3.33 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro	+138.5	Gas-Wind	-62.1
Solar-Wind	-136.0	Nuclear-Solar	-61.6
Bioenergy-OFos	+122.0	Coal-Solar	-61.2
Bioenergy-Gas	+115.6	Hydro-OFos	+61.0
Bioenergy-Nuclear	+114.4	Nuclear-Wind	-60.7
Bioenergy-Coal	+114.0	Coal-Wind	-60.3
Bioenergy-Wind	-104.3	Gas-Nuclear	+46.6
Bioenergy-Solar	-103.8	Coal-Nuclear	+44.5
Hydro-Solar	-77.8	Coal-Gas	+42.9
Hydro-Wind	-77.3	Gas-OFos	+37.5
OFos-Solar	-68.6	Nuclear-OFos	+36.0
Hydro-Nuclear	+68.1	Coal-OFos	+35.5
OFos-Wind	-67.9	Gas-Hydro	+23.0
Gas-Solar	-62.9	Coal-Hydro	+21.5

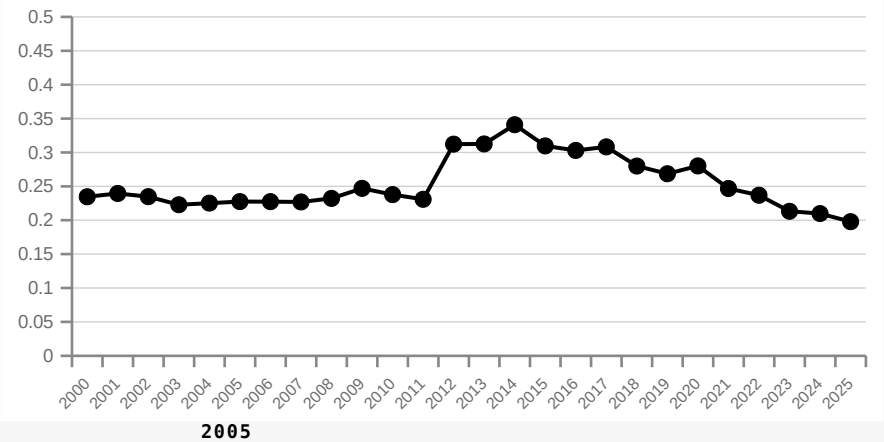
METRIC LEDGER

```
Hs     = 0.225250   Ring = Hs-2
E_metric = 37.663052   kappa = 232.59
omega   = 3.3324 deg
d_A     = 0.582491
Helm    = Wind delta = +0.435321

CLR:
 Bio -0.862 Coal +1.933 Gas +1.797 Hyd +0.763
 Nuc +1.900 OFos +1.378 Sol -3.517 Wind -3.391

Steering (1/x_j):
 Bio  62.1 Coal  3.8 Gas   4.3 Hyd  12.2
 Nuc   3.9 OFos  6.6 Sol 882.8 Wind 778.6
```

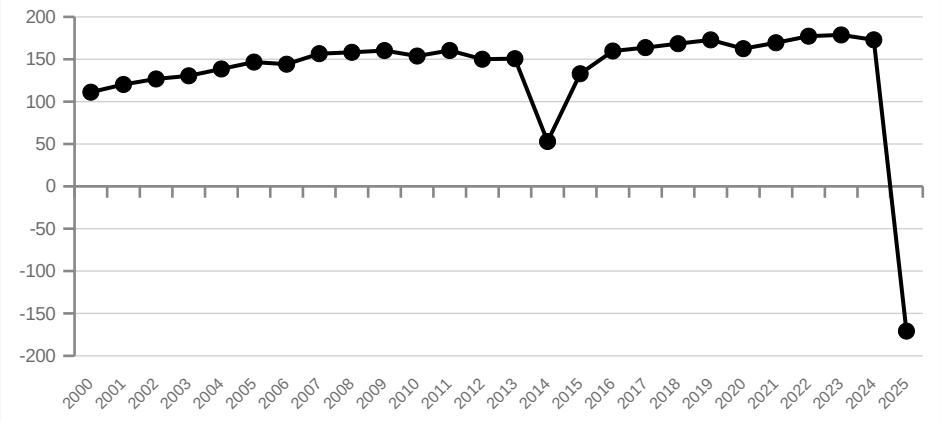
SECTION PLATE t=5 | Year 2005

Hs = 0.2276 | Hs-2 | omega = 2.48 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro	+146.7	Gas-Solar	-63.3
Solar-Wind	-136.4	Gas-Wind	-62.2
Bioenergy-OFos	+119.2	Nuclear-Solar	-61.3
Bioenergy-Gas	+114.5	Coal-Solar	-60.2
Bioenergy-Nuclear	+112.7	Nuclear-Wind	-60.2
Bioenergy-Coal	+111.8	Coal-Wind	-59.0
Bioenergy-Wind	-103.5	Gas-Nuclear	+47.4
Bioenergy-Solar	-102.9	Coal-Nuclear	+43.7
Hydro-Solar	-81.4	Coal-Gas	+41.3
Hydro-Wind	-81.0	Gas-OFos	+39.2
Hydro-Nuclear	+74.6	Nuclear-OFos	+36.8
Hydro-OFos	+69.8	Coal-OFos	+35.6
OFos-Solar	-67.7	Gas-Hydro	+16.7
OFos-Wind	-66.7	Coal-Hydro	+14.8

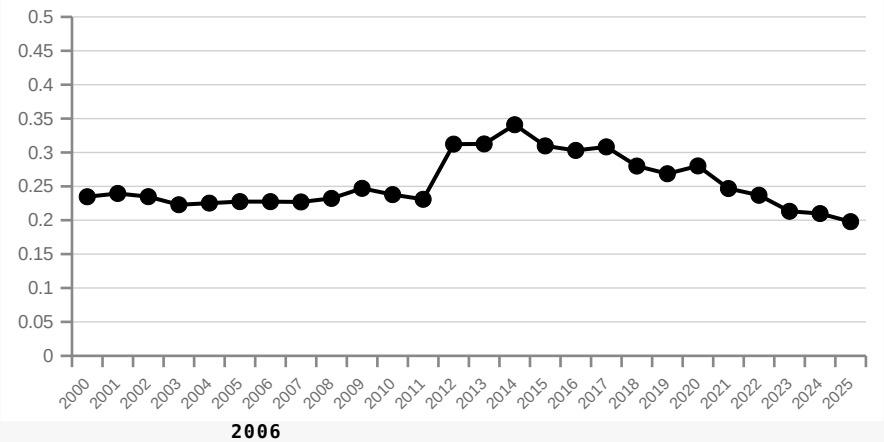
METRIC LEDGER

```
Hs     = 0.227556   Ring = Hs-2
E_metric = 34.138859   kappa = 195.83
omega   = 2.4787 deg
d_A     = 0.391966
Helm    = Hyd delta = -0.256970

CLR:
 Bio -0.769 Coal +1.920 Gas +1.688 Hyd +0.506
 Nuc +1.835 OFos +1.375 Sol -3.357 Wind -3.198

Steering (1/x_j):
 Bio   53.2 Coal   3.6 Gas   4.6 Hyd   14.9
 Nuc    3.9 OFos   6.2 Sol  707.4 Wind 603.7
```

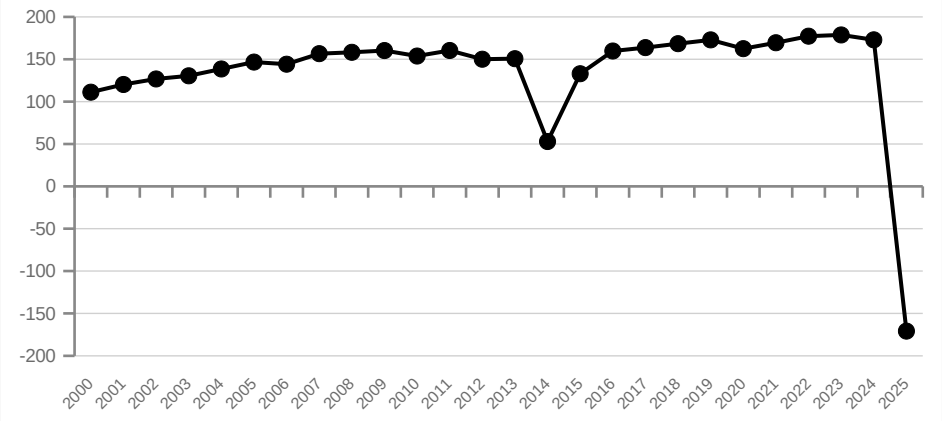
SECTION PLATE t=6 | Year 2006

Hs = 0.2275 | Hs-2 | omega = 2.66 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro +144.2	Gas-Solar -61.4
Solar-Wind -135.6	Gas-Wind -60.9
Bioenergy-OFos +125.7	Nuclear-Solar -60.3
Bioenergy-Gas +115.3	Coal-Solar -60.0
Bioenergy-Nuclear +114.3	Nuclear-Wind -59.8
Bioenergy-Coal +114.0	Coal-Wind -59.5
Bioenergy-Wind -104.7	Gas-Nuclear +46.3
Bioenergy-Solar -104.4	Coal-Nuclear +44.6
Hydro-Solar -79.5	Coal-Gas +43.3
Hydro-Wind -79.3	Gas-OFos +33.3
Hydro-Nuclear +72.0	Nuclear-OFos +32.1
OFos-Solar -70.3	Coal-OFos +31.8
OFos-Wind -69.9	Gas-Hydro +18.8
Hydro-OFos +62.6	Coal-Hydro +17.8

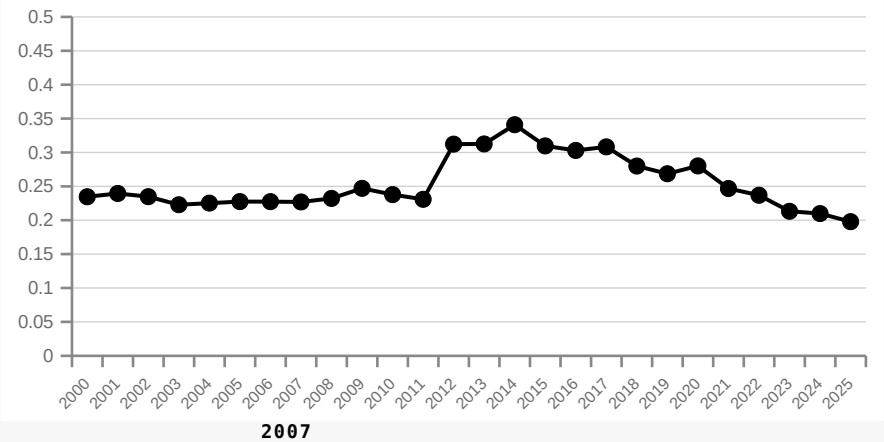
METRIC LEDGER

```
Hs    = 0.227477   Ring = Hs-2
E_metric = 32.200850   kappa = 155.67
omega   = 2.6642 deg
d_A    = 0.316211
Helm    = OFos delta = -0.228493

CLR:
 Bio -0.823  Coal +1.849  Gas +1.743  Hyd +0.594
 Nuc +1.826  OFos +1.146  Sol -3.199  Wind -3.136

Steering (1/x_j):
 Bio   54.1 Coal   3.7 Gas   4.2 Hyd   13.1
 Nuc    3.8 OFos   7.6 Sol  582.2 Wind  546.6
```

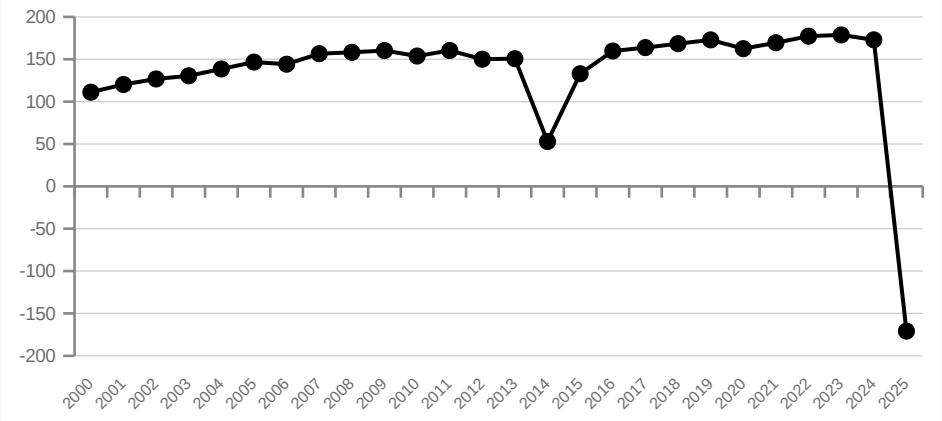
SECTION PLATE t=7 | Year 2007

Hs = 0.2270 | Hs-2 | omega = 3.21 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro +156.6	Nuclear-Solar -61.5
Solar-Wind -136.6	Gas-Solar -60.6
Bioenergy-OFos +123.5	Nuclear-Wind -60.1
Bioenergy-Nuclear +116.6	Coal-Solar -59.7
Bioenergy-Gas +115.8	Gas-Wind -59.2
Bioenergy-Coal +115.0	Coal-Wind -58.3
Bioenergy-Wind -106.1	Gas-Nuclear +44.0
Bioenergy-Solar -105.2	Coal-Gas +43.9
Hydro-Solar -83.3	Coal-Nuclear +42.9
Hydro-Wind -82.9	Nuclear-OFos +37.1
Hydro-Nuclear +77.8	Gas-OFos +36.1
Hydro-OFos +74.0	Coal-OFos +35.1
OFos-Solar -67.7	Gas-Hydro +11.8
OFos-Wind -66.5	Coal-Hydro +11.4

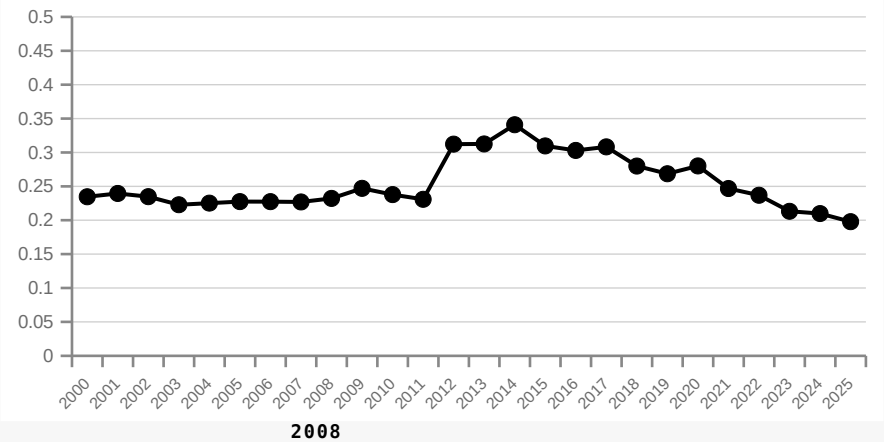
METRIC LEDGER

```
Hs     = 0.227024   Ring = Hs-2
E_metric = 29.954647   kappa = 137.36
omega   = 3.2122 deg
d_A     = 0.371740
Helm    = Hyd delta = -0.228639

CLR:
 Bio -0.845  Coal +1.815  Gas +1.748  Hyd +0.366
 Nuc +1.686  OFos +1.276  Sol -3.108  Wind -2.937

Steering (1/x_j):
 Bio   53.2 Coal   3.7 Gas   4.0 Hyd   15.8
 Nuc    4.2 OFos   6.4 Sol  510.9 Wind  430.7
```

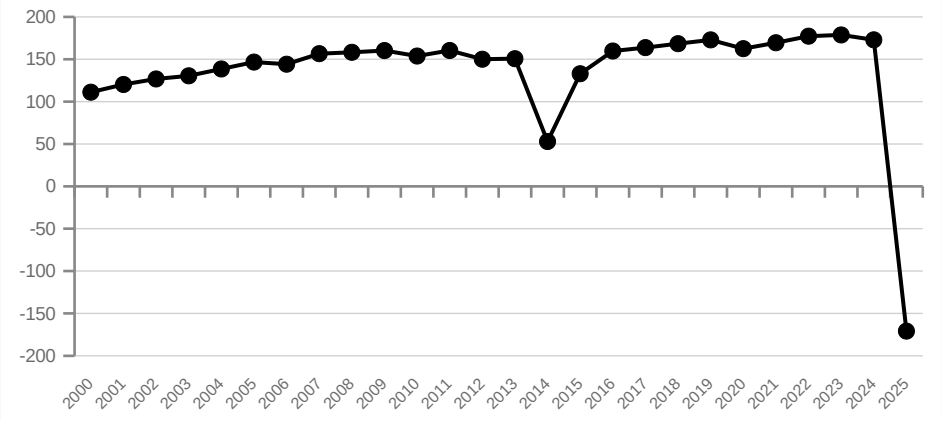
SECTION PLATE t=8 | Year 2008

Hs = 0.2323 | Hs-2 | omega = 2.01 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro	+158.2	Nuclear-Solar	-62.4
Solar-Wind	-136.2	Nuclear-Wind	-61.4
Bioenergy-OFos	+127.2	Gas-Solar	-58.8
Bioenergy-Nuclear	+119.9	Coal-Solar	-58.4
Bioenergy-Gas	+116.3	Gas-Wind	-57.7
Bioenergy-Coal	+116.0	Coal-Wind	-57.3
Bioenergy-Wind	-107.4	Coal-Gas	+44.6
Bioenergy-Solar	-106.7	Gas-Nuclear	+40.7
Hydro-Solar	-83.2	Coal-Nuclear	+40.3
Hydro-Wind	-82.9	Nuclear-OFos	+37.1
Hydro-Nuclear	+77.1	Gas-OFos	+33.0
Hydro-OFos	+73.1	Coal-OFos	+32.6
OFos-Solar	-68.5	Gas-Hydro	+11.2
OFos-Wind	-67.6	Coal-Hydro	+11.0

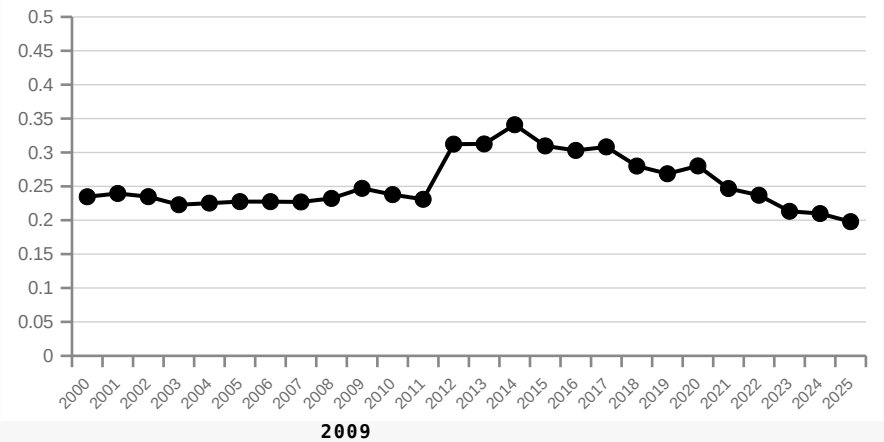
METRIC LEDGER

```
Hs    = 0.232256   Ring = Hs-2
E_metric = 28.906425   kappa = 128.84
omega   = 2.0079 deg
d_A    = 0.213233
Helm    = Nuc delta = -0.116881

CLR:
 Bio -0.901  Coal +1.851  Gas +1.824  Hyd +0.360
 Nuc +1.569  OFos +1.185  Sol -3.008  Wind -2.881

Steering (1/x_j):
 Bio   55.6 Coal   3.5 Gas   3.6 Hyd   15.7
 Nuc    4.7 OFos   6.9 Sol  457.0 Wind  402.6
```

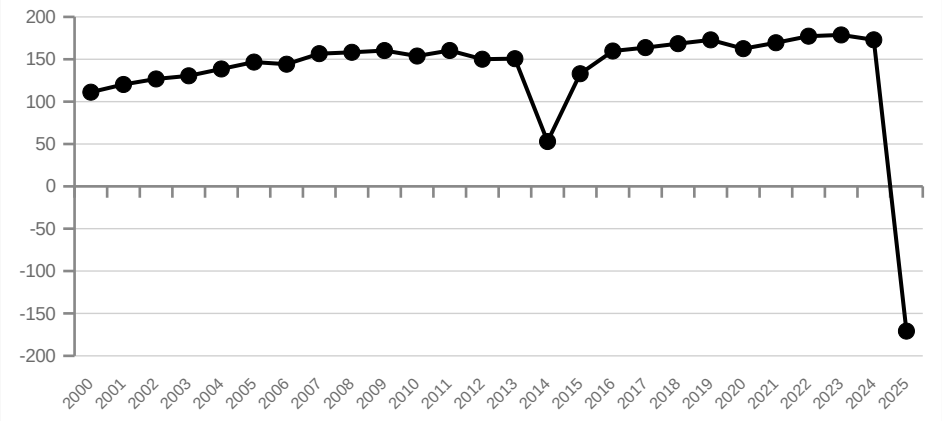
SECTION PLATE t=9 | Year 2009

Hs = 0.2471 | Hs-2 | omega = 5.28 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro +160.2	Nuclear-Solar -59.0
Bioenergy-OFos +141.4	Nuclear-Wind -57.9
Solar-Wind -136.2	Coal-Solar -57.1
Bioenergy-Nuclear +118.4	Gas-Solar -56.6
Bioenergy-Coal +116.7	Coal-Wind -56.0
Bioenergy-Gas +116.2	Gas-Wind -55.4
Bioenergy-Wind -108.8	Coal-Gas +45.6
Bioenergy-Solar -108.0	Coal-Nuclear +42.9
Hydro-Solar -83.3	Gas-Nuclear +42.3
Hydro-Wind -83.0	Nuclear-OFos +23.4
Hydro-Nuclear +79.0	Coal-OFos +21.9
OFos-Solar -75.4	Gas-OFos +21.5
OFos-Wind -74.8	Coal-Hydro +10.3
Hydro-OFos +65.8	Gas-Hydro +10.0

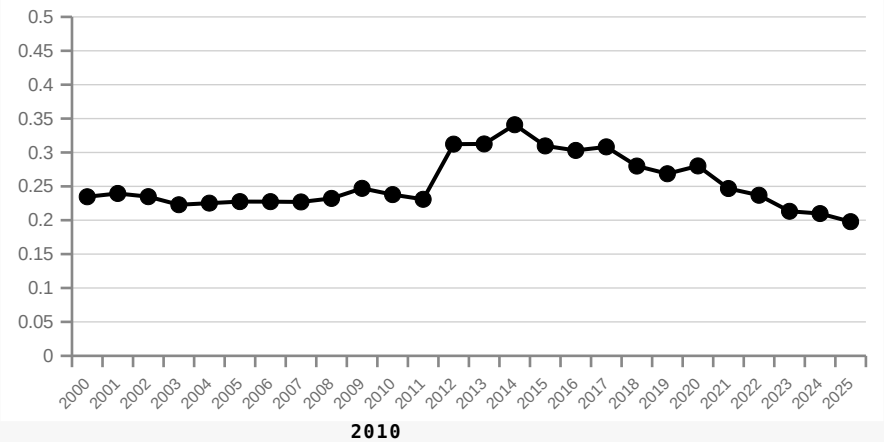
METRIC LEDGER

```
Hs    = 0.247095   Ring = Hs-2
E_metric = 26.238217   kappa = 106.47
omega   = 5.2774 deg
d_A    = 0.545961
Helm    = OFos delta = -0.454982

CLR:
 Bio -0.915 Coal +1.817 Gas +1.856 Hyd +0.329
 Nuc +1.689 OFos +0.730 Sol -2.812 Wind -2.694

Steering (1/x_j):
 Bio   54.8 Coal   3.6 Gas   3.4 Hyd   15.8
 Nuc    4.1 OFos  10.6 Sol  365.3 Wind  324.8
```

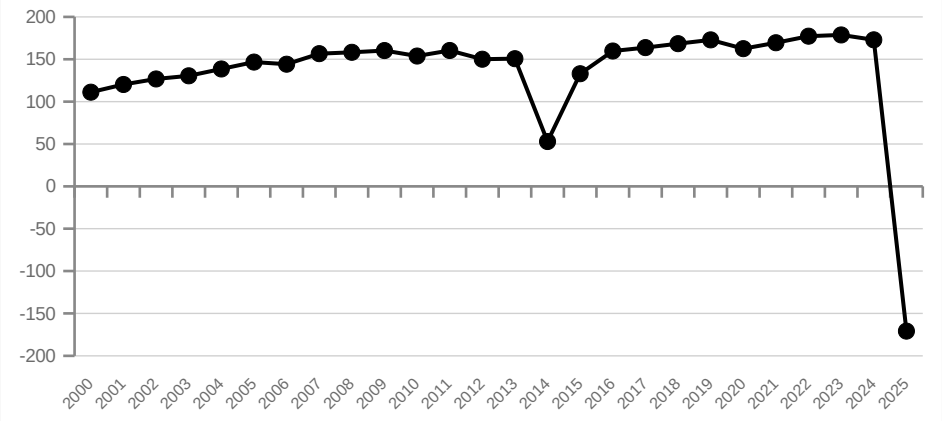
SECTION PLATE t=10 | Year 2010

Hs = 0.2379 | Hs-2 | omega = 2.40 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro +153.8	Nuclear-Solar -57.9
Bioenergy-OFos +147.0	Coal-Wind -56.8
Solar-Wind -134.9	Coal-Solar -56.7
Bioenergy-Nuclear +119.5	Gas-Wind -56.3
Bioenergy-Coal +118.4	Gas-Solar -56.2
Bioenergy-Gas +118.0	Hydro-OFos +52.8
Bioenergy-Solar -109.6	Coal-Gas +45.5
Bioenergy-Wind -109.5	Coal-Nuclear +43.6
Hydro-Wind -80.1	Gas-Nuclear +43.2
Hydro-Solar -80.1	Nuclear-OFos +20.2
OFos-Wind -77.1	Coal-OFos +19.3
OFos-Solar -77.0	Gas-OFos +19.0
Hydro-Nuclear +74.4	Coal-Hydro +14.9
Nuclear-Wind -58.0	Gas-Hydro +14.7

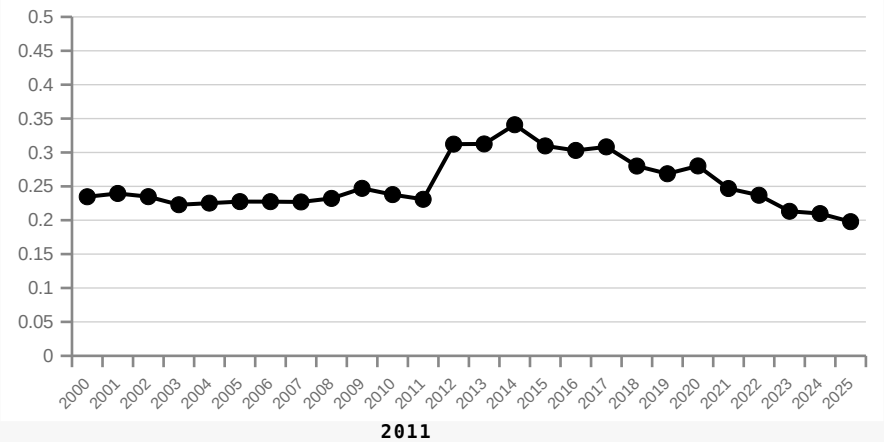
METRIC LEDGER

```
Hs     = 0.237884   Ring = Hs-2
E_metric = 24.350226   kappa = 82.97
omega   = 2.4032 deg
d_A     = 0.282318
Helm    = Sol delta = +0.171611

CLR:
 Bio -0.938  Coal +1.737  Gas +1.766  Hyd +0.462
 Nuc +1.657  OFos +0.609  Sol -2.640  Wind -2.653

Steering (1/x_j):
 Bio   53.0  Coal   3.6  Gas    3.5  Hyd   13.1
 Nuc    4.0  OFos  11.3  Sol  290.5  Wind 294.2
```

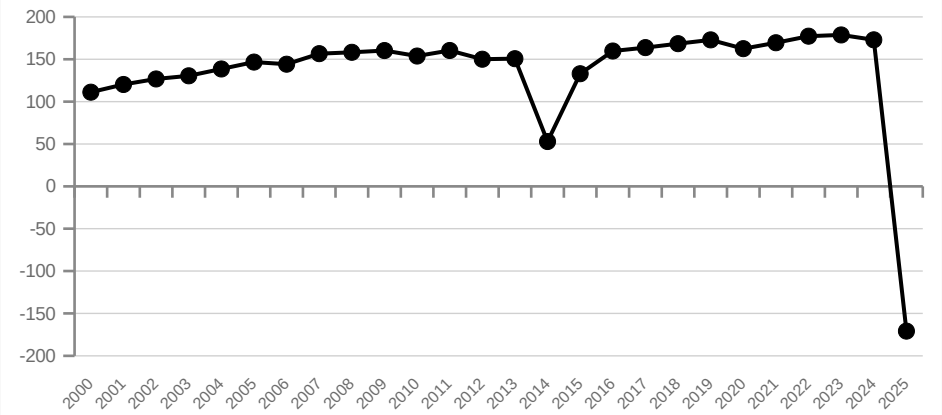
SECTION PLATE t=11 | Year 2011

Hs = 0.2309 | Hs-2 | omega = 9.40 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro +160.3	OFos-Solar -66.2
Bioenergy-OFos +134.0	Nuclear-Solar -66.2
Bioenergy-Nuclear +134.0	Coal-Wind -58.0
Solar-Wind -132.7	Coal-Solar -55.9
Bioenergy-Coal +122.1	Gas-Wind -53.6
Bioenergy-Gas +118.0	Gas-Solar -51.4
Bioenergy-Solar -113.0	Coal-Gas +49.7
Bioenergy-Wind -111.4	Nuclear-OFos +45.0
Hydro-Wind -82.0	Coal-Nuclear +33.0
Hydro-Solar -81.4	Coal-OFos +33.0
Hydro-Nuclear +71.0	Gas-Nuclear +28.9
Hydro-OFos +70.9	Gas-OFos +28.9
OFos-Wind -67.9	Coal-Hydro +12.7
Nuclear-Wind -67.9	Gas-Hydro +10.8

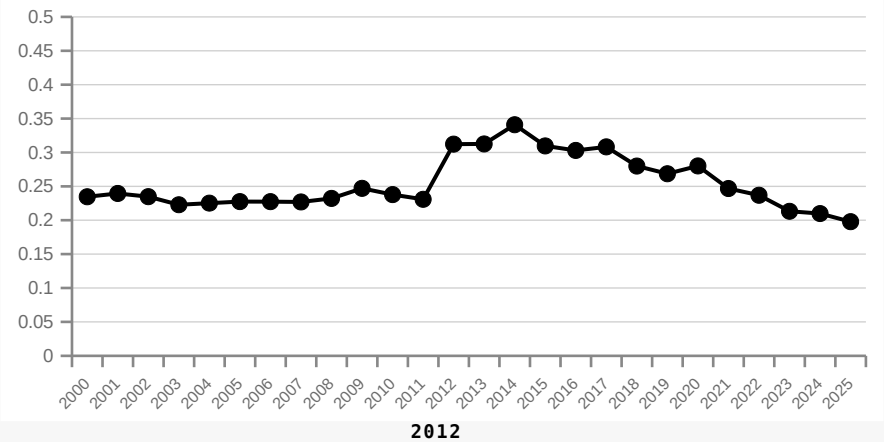
METRIC LEDGER

```
Hs     = 0.230931   Ring = Hs-2
E_metric = 21.527052   kappa = 85.16
omega   = 9.4013 deg
d_A     = 0.837843
Helm    = Nuc delta = -0.617218

CLR:
 Bio -1.004  Coal +1.599  Gas +1.886  Hyd +0.359
 Nuc +1.040  OFos +1.039  Sol -2.360  Wind -2.559

Steering (1/x_j):
 Bio   52.3 Coal   3.9 Gas   2.9 Hyd   13.4
 Nuc    6.8 OFos   6.8 Sol  203.0 Wind  247.6
```

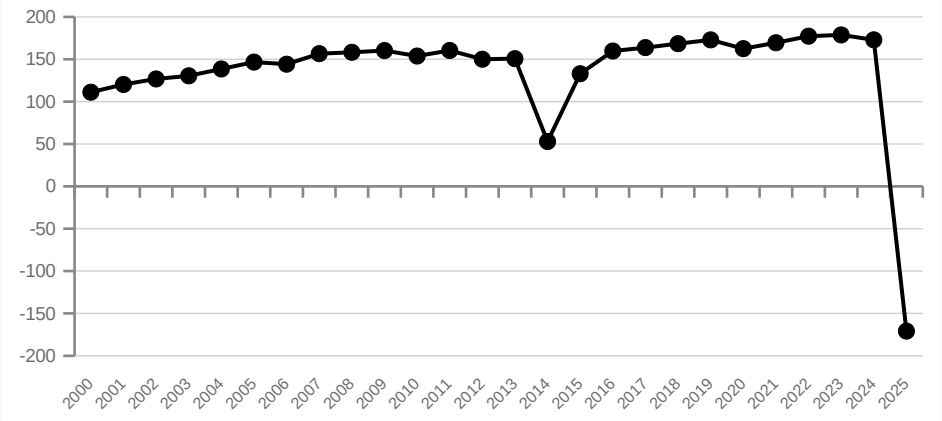
SECTION PLATE t=12 | Year 2012

Hs = 0.3123 | Hs-3 | omega = 27.46 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro	+150.0	Hydro-Nuclear	-65.4
Solar-Wind	-129.0	OFos-Wind	-58.2
Bioenergy-Nuclear	-128.4	OFos-Solar	-52.6
Nuclear-OFos	+124.6	Coal-Wind	-50.5
Bioenergy-OFos	+118.7	Coal-Gas	+48.7
Nuclear-Solar	-117.8	Gas-Wind	-46.9
Nuclear-Wind	-113.2	Coal-Solar	-44.5
Bioenergy-Solar	-112.7	Gas-Solar	-40.8
Bioenergy-Coal	+112.4	Coal-OFos	+37.0
Bioenergy-Gas	+109.9	Gas-OFos	+33.5
Bioenergy-Wind	-108.7	Coal-Nuclear	-27.4
Hydro-Wind	-78.9	Gas-Nuclear	-24.5
Hydro-Solar	-76.4	Coal-Hydro	+13.4
Hydro-OFos	+72.5	Gas-Hydro	+11.8

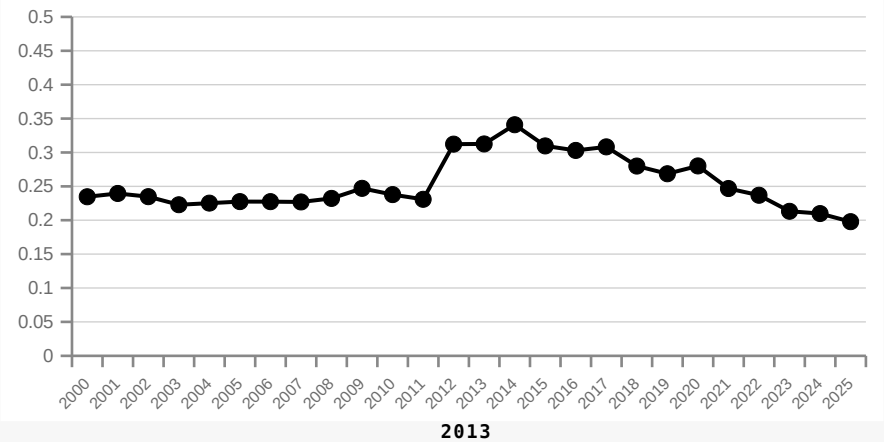
METRIC LEDGER

```
Hs     = 0.312291   Ring = Hs-3
E_metric = 21.442593   kappa = 92.07
omega   = 27.4642 deg
d_A     = 2.200635
Helm    = Nuc delta = -2.038262

CLR:
 Bio -0.792  Coal +1.924  Gas +2.188  Hyd +0.457
 Nuc -0.999  OFos +1.448  Sol -1.891  Wind -2.335

Steering (1/x_j):
 Bio   50.0 Coal   3.3 Gas    2.5 Hyd   14.4
 Nuc   61.5 OFos   5.3 Sol  150.2 Wind  234.0
```

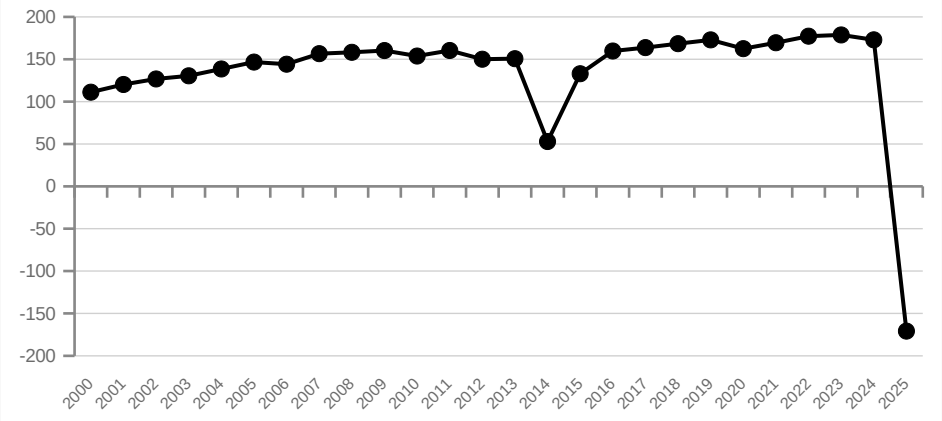
SECTION PLATE t=13 | Year 2013

Hs = 0.3126 | Hs-3 | omega = 7.54 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro	+150.7	Hydro-OFos	+69.5
Nuclear-OFos	+136.6	OFos-Wind	-62.7
Nuclear-Solar	-132.3	Coal-Wind	-49.5
Bioenergy-OFos	+123.6	OFos-Solar	-49.2
Bioenergy-Nuclear	-122.2	Gas-Wind	-47.3
Solar-Wind	-120.9	Coal-Gas	+47.2
Bioenergy-Solar	-119.8	Coal-Solar	-35.0
Nuclear-Wind	-118.6	Gas-Solar	-33.0
Bioenergy-Coal	+111.9	Coal-Nuclear	-32.5
Bioenergy-Gas	+110.4	Coal-OFos	+31.1
Bioenergy-Wind	-108.9	Gas-Nuclear	-30.6
Hydro-Wind	-79.1	Gas-OFos	+29.2
Hydro-Solar	-72.1	Coal-Hydro	+12.7
Hydro-Nuclear	-70.5	Gas-Hydro	+11.8

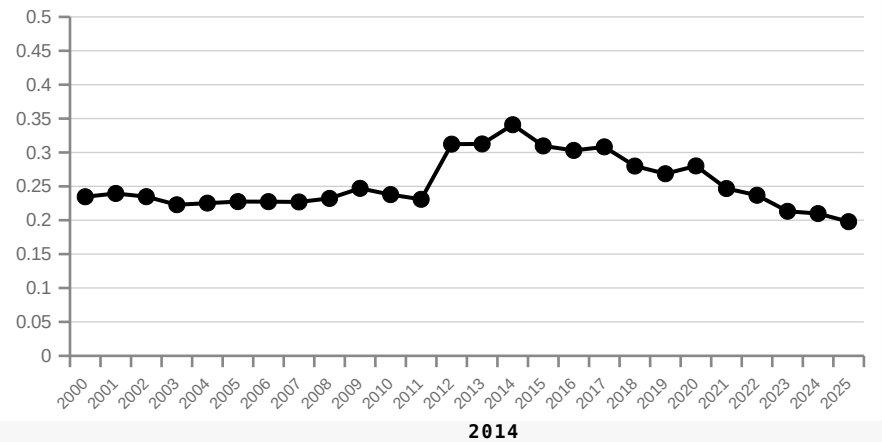
METRIC LEDGER

```
Hs     = 0.312583    Ring = Hs-3
E_metric = 19.271571    kappa = 82.57
omega    = 7.5396 deg
d_A     = 0.639857
Helm     = Sol delta = +0.518125

CLR:
 Bio -0.788  Coal +1.962  Gas +2.118  Hyd +0.443
 Nuc -1.250  OFos +1.184  Sol -1.373  Wind -2.296

Steering (1/x_j):
 Bio   46.9 Coal   3.0 Gas    2.6 Hyd   13.7
 Nuc   74.5 OFos   6.5 Sol   84.3 Wind 212.0
```

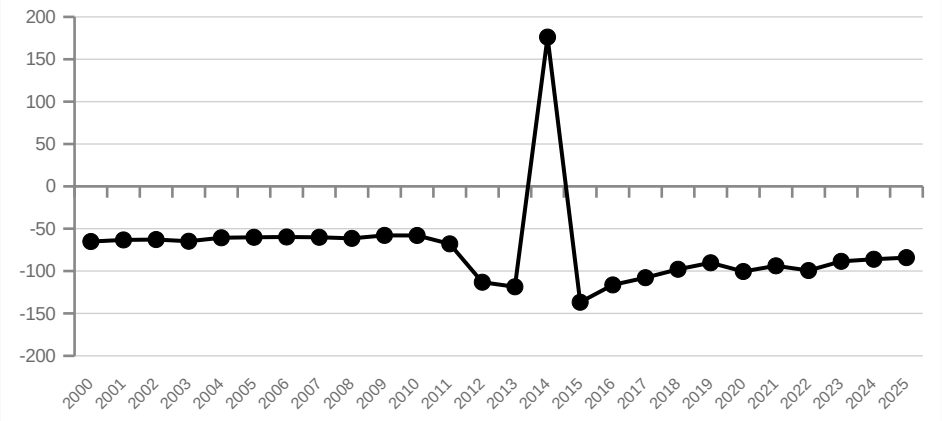
SECTION PLATE t=14 | Year 2014

Hs = 0.3409 | Hs-3 | omega = 66.11 deg

YZ: Hs vs Time



XZ: Bearing (Nucl-Wind) vs Time



XY PLAN VIEW: All Bearings (deg)

Nuclear-Wind +176.1	Coal-OFos +39.9
Nuclear-Solar +173.5	Gas-OFos +38.9
Nuclear-OFos +170.8	Coal-Hydro +37.8
Bioenergy-Nuclear -83.5	Hydro-Solar +37.1
Hydro-Nuclear -81.5	Gas-Hydro +36.8
Coal-Nuclear -79.1	OFos-Solar +35.0
Gas-Nuclear -78.7	Solar-Wind +30.8
Bioenergy-Gas +60.5	Bioenergy-Wind +30.8
Bioenergy-Coal +59.6	Coal-Solar +30.4
Bioenergy-OFos +55.0	Gas-Solar +29.5
Bioenergy-Hydro +52.9	Hydro-Wind +24.2
Hydro-OFos +47.2	OFos-Wind +22.6
Coal-Gas +46.0	Coal-Wind +19.3
Bioenergy-Solar +45.0	Gas-Wind +18.6

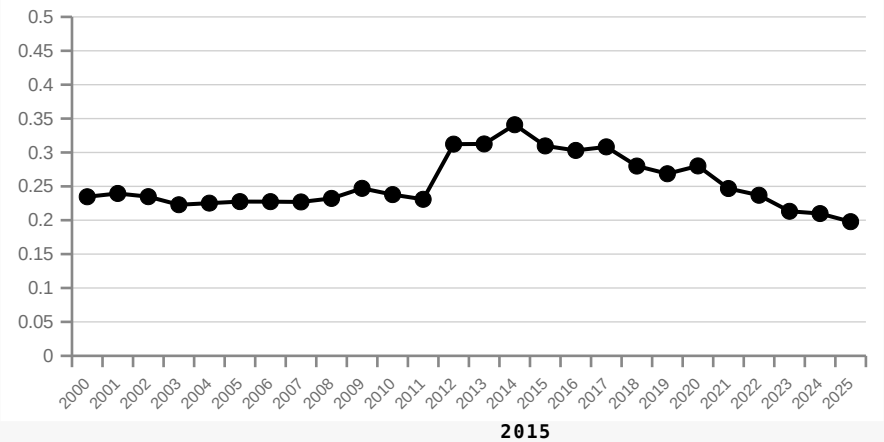
METRIC LEDGER

```
Hs    = 0.340926   Ring = Hs-3
E_metric = 896.732001   kappa = 4.5e+17
omega   = 66.1070 deg
d_A    = 34.870573
Helm    = Nuc delta = -32.612482

CLR:
 Bio +3.839  Coal +6.550  Gas +6.788  Hyd +5.080
 Nuc -33.862  OFos +5.483  Sol +3.835  Wind +2.288

Steering (1/x_j):
 Bio   45.0 Coal   3.0 Gas   2.4 Hyd   13.0
 Nuc  1e+15 OFos   8.7 Sol  45.1 Wind 212.1
```

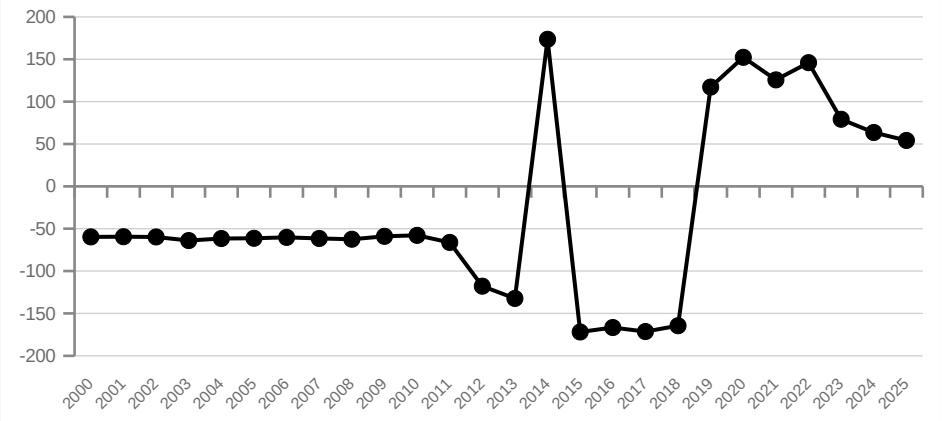
SECTION PLATE t=15 | Year 2015

Hs = 0.3097 | Hs-3 | omega = 49.67 deg

YZ: Hs vs Time



XZ: Bearing (Nucl-Sola) vs Time



XY PLAN VIEW: All Bearings (deg)

Nuclear-Solar	-171.9	Hydro-OFos	+53.4
Nuclear-OFos	+162.1	Coal-Nuclear	-50.2
Bioenergy-Solar	-147.5	Coal-Wind	-48.4
Nuclear-Wind	-136.8	Gas-Nuclear	-47.8
Bioenergy-Hydro	+132.9	Coal-Gas	+47.5
Bioenergy-OFos	+124.7	Gas-Wind	-46.0
Bioenergy-Coal	+105.0	Hydro-Solar	-30.6
Bioenergy-Gas	+103.8	OFos-Solar	-23.8
Bioenergy-Wind	-103.4	Coal-OFos	+21.2
Bioenergy-Nuclear	-102.6	Gas-OFos	+19.6
Solar-Wind	-98.6	Coal-Hydro	+16.1
Hydro-Nuclear	-76.5	Gas-Hydro	+14.9
Hydro-Wind	-75.6	Coal-Solar	-9.7
OFos-Wind	-71.0	Gas-Solar	-8.9

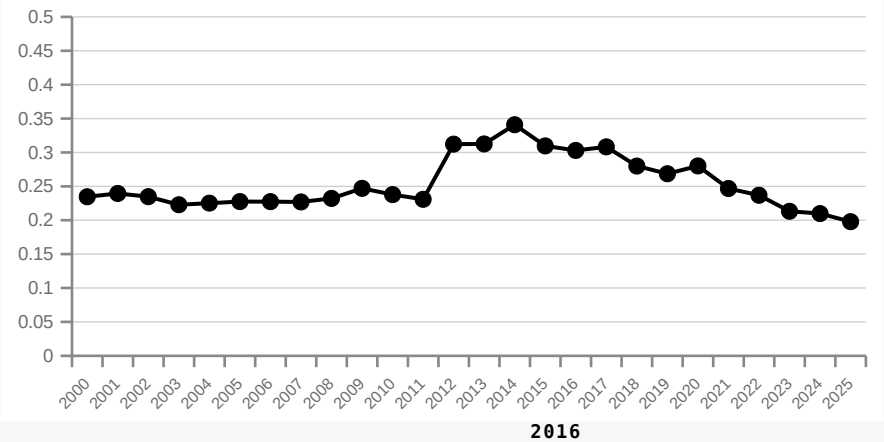
METRIC LEDGER

```
Hs     = 0.309676    Ring = Hs-3
E_metric = 20.447922    kappa = 92.37
omega    = 49.6746 deg
d_A     = 33.667556
Helm     = Nuc delta = +31.490431

CLR:
 Bio -0.531  Coal +1.976  Gas +2.154  Hyd +0.571
 Nuc -2.372  OFos +0.768  Sol -0.338  Wind -2.228

Steering (1/x_j):
 Bio   36.2 Coal   2.9 Gas    2.5 Hyd   12.0
 Nuc  227.9 OFos   9.9 Sol   29.8 Wind  197.3
```

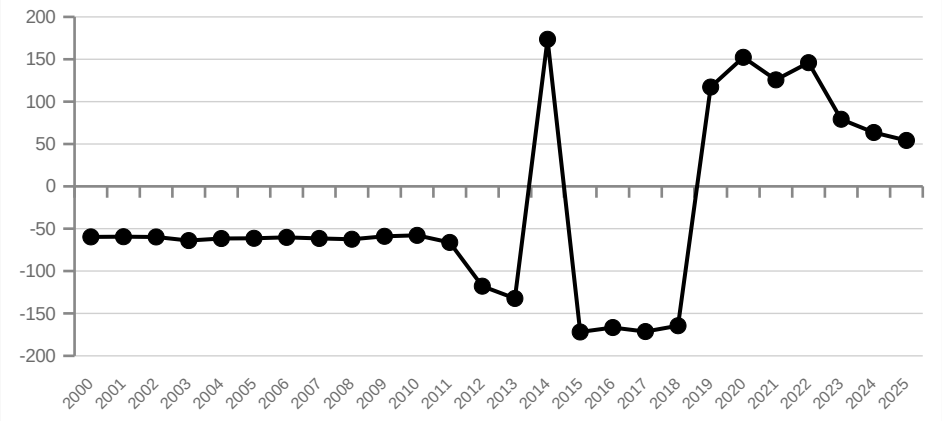
SECTION PLATE t=16 | Year 2016

Hs = 0.3030 | Hs-3 | omega = 16.26 deg

YZ: Hs vs Time



XZ: Bearing (Nucl-Sola) vs Time



XY PLAN VIEW: All Bearings (deg)

Nuclear-Solar	-166.6	Hydro-OFos	+58.3
Bioenergy-Solar	-162.5	Coal-Wind	-53.2
Bioenergy-Hydro	+159.7	Coal-Gas	+49.5
Nuclear-OFos	+155.8	Gas-Wind	-48.8
Bioenergy-OFos	+149.1	Hydro-Solar	-40.5
Bioenergy-Nuclear	-126.9	Coal-Nuclear	-33.5
Bioenergy-Coal	+116.5	Gas-Nuclear	-29.5
Nuclear-Wind	-116.4	OFos-Solar	-27.8
Bioenergy-Gas	+113.0	Coal-OFos	+16.6
Bioenergy-Wind	-110.4	Gas-OFos	+14.3
Solar-Wind	-96.7	Coal-Hydro	+10.4
Hydro-Wind	-82.2	Coal-Solar	-8.9
OFos-Wind	-77.4	Gas-Hydro	+8.9
Hydro-Nuclear	-74.5	Gas-Solar	-7.6

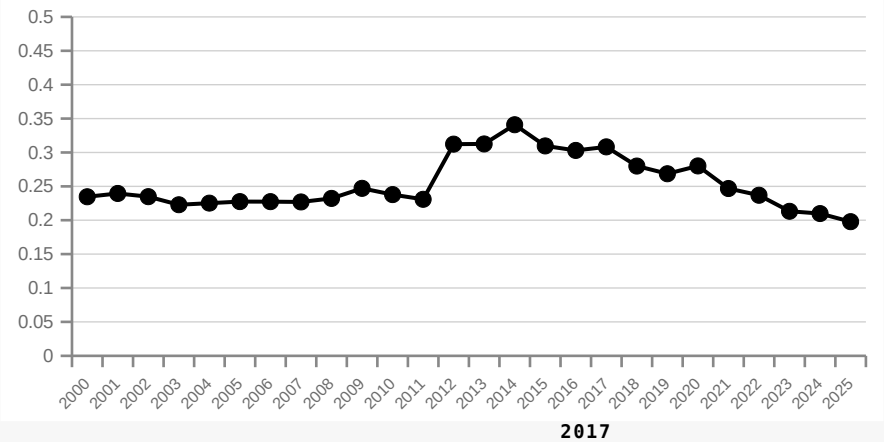
METRIC LEDGER

```
Hs     = 0.303033   Ring = Hs-3
E_metric = 15.740383   kappa = 85.85
omega    = 16.2591 deg
d_A    = 1.320048
Helm    = Nuc delta = +1.196283

CLR:
 Bio -0.884  Coal +1.777  Gas +2.080  Hyd +0.327
 Nuc -1.176  OFos +0.529  Sol -0.279  Wind -2.373

Steering (1/x_j):
 Bio   44.9 Coal   3.1 Gas   2.3 Hyd   13.4
 Nuc   60.2 OFos  10.9 Sol  24.5 Wind 199.2
```

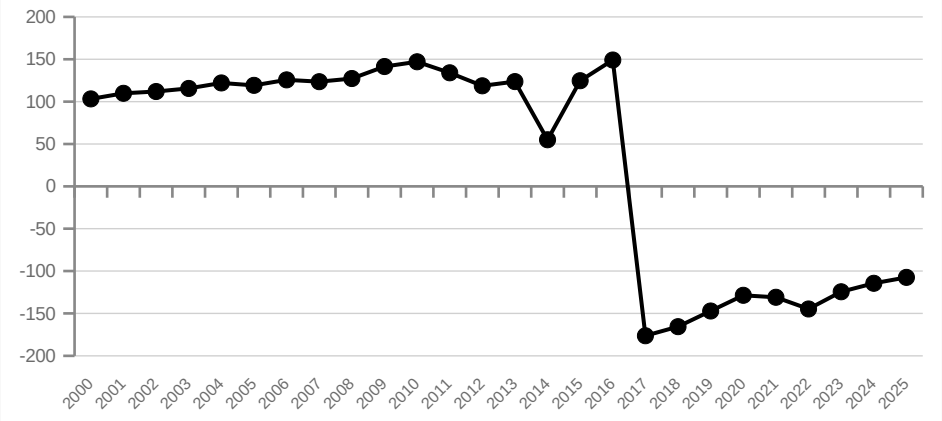
SECTION PLATE t=17 | Year 2017

Hs = 0.3083 | Hs-3 | omega = 10.95 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-OFos) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-OFos	-176.3	Hydro-Nuclear	-69.6
Nuclear-OFos	-175.3	Coal-Wind	-51.7
Bioenergy-Solar	-173.2	Coal-Gas	+48.6
Nuclear-Solar	-171.4	Gas-Wind	-48.1
Bioenergy-Hydro	+163.7	Hydro-Solar	-22.2
Bioenergy-Nuclear	-141.9	Coal-Nuclear	-22.2
OFos-Solar	-118.2	Gas-Nuclear	-19.7
Bioenergy-Coal	+117.4	Hydro-OFos	-12.4
Bioenergy-Gas	+114.5	Coal-Hydro	+8.6
Bioenergy-Wind	-112.3	Gas-Hydro	+7.6
Nuclear-Wind	-107.8	Coal-Solar	-3.5
Solar-Wind	-92.8	Gas-Solar	-3.1
OFos-Wind	-91.5	Coal-OFos	-1.9
Hydro-Wind	-83.2	Gas-OFos	-1.7

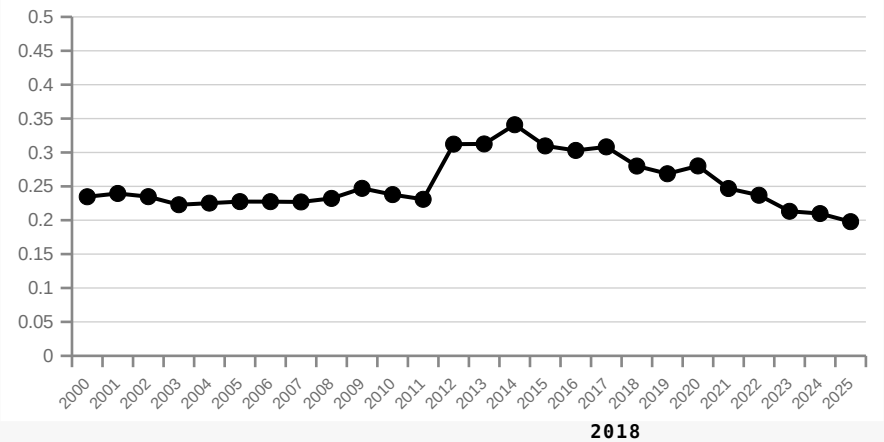
METRIC LEDGER

```
Hs     = 0.308283    Ring = Hs-3
E_metric = 14.038510    kappa = 74.62
omega    = 10.9468 deg
d_A     = 0.767882
Helm    = OFos delta = -0.588595

CLR:
 Bio -0.931  Coal +1.795  Gas +2.039  Hyd +0.272
 Nuc -0.731  OFos -0.060  Sol -0.111  Wind -2.273

Steering (1/x_j):
 Bio   45.2 Coal   3.0 Gas    2.3 Hyd   13.6
 Nuc   37.0 OFos  18.9 Sol   19.9 Wind  173.1
```

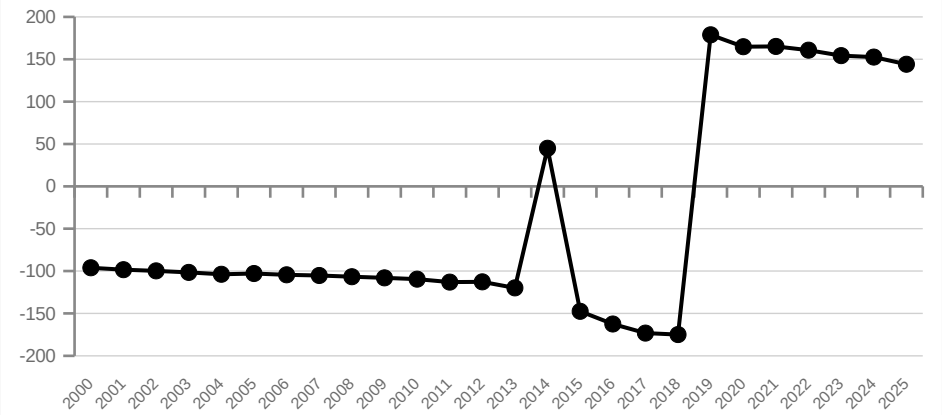
SECTION PLATE t=18 | Year 2018

Hs = 0.2800 | Hs-2 | omega = 7.20 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Sola) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Solar	-174.9	Hydro-Nuclear	-57.3
Bioenergy-Hydro	+168.4	Coal-Wind	-52.5
Bioenergy-OFos	-165.6	Hydro-OFos	-51.3
Nuclear-Solar	-164.5	Gas-Wind	-49.2
Bioenergy-Nuclear	-162.2	Coal-Gas	+48.4
OFos-Solar	-160.9	Hydro-Solar	-23.4
Nuclear-OFos	-141.4	Coal-Nuclear	-10.2
Bioenergy-Coal	+119.4	Gas-Nuclear	-9.1
Bioenergy-Gas	+116.6	Coal-OFos	-8.2
Bioenergy-Wind	-113.4	Gas-OFos	-7.3
Nuclear-Wind	-97.9	Coal-Hydro	+6.6
OFos-Wind	-96.3	Gas-Hydro	+5.9
Solar-Wind	-92.2	Coal-Solar	-2.9
Hydro-Wind	-84.9	Gas-Solar	-2.5

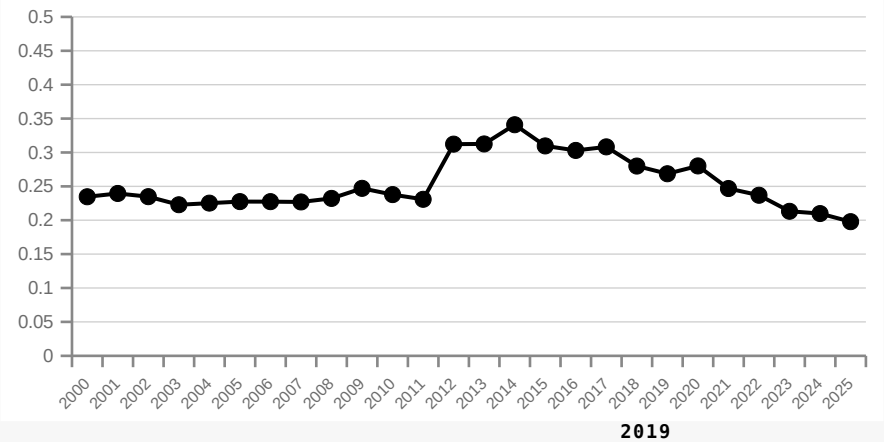
METRIC LEDGER

```
Hs    = 0.279970   Ring = Hs-2
E_metric = 12.412465   kappa = 60.46
omega   = 7.1966 deg
d_A    = 0.507945
Helm    = Nuc delta = +0.426213

CLR:
 Bio -0.951  Coal +1.689  Gas +1.901  Hyd +0.196
 Nuc -0.305  OFos -0.244  Sol -0.084  Wind -2.202

Steering (1/x_j):
 Bio   42.1 Coal   3.0 Gas   2.4 Hyd   13.4
 Nuc   22.1 OFos  20.7 Sol  17.7 Wind 146.9
```

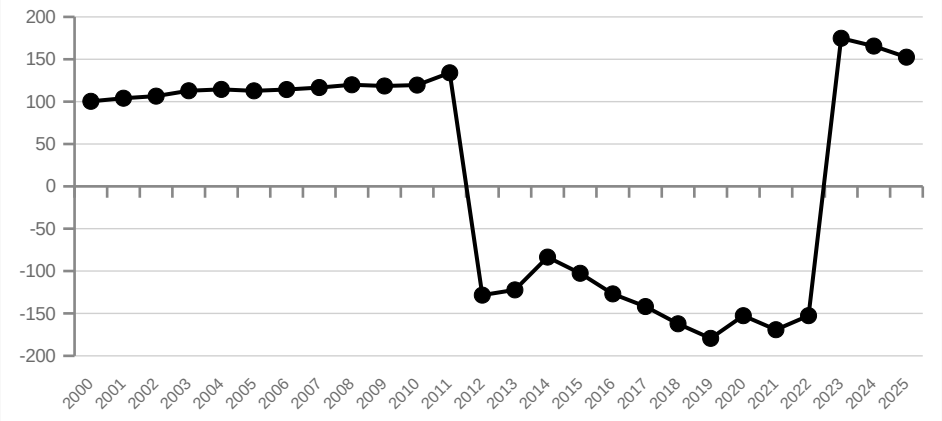
SECTION PLATE t=19 | Year 2019

Hs = 0.2686 | Hs-2 | omega = 7.62 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Nucl) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Nuclear -179.4	Hydro-OFos -79.0
Bioenergy-Solar +178.8	Coal-Wind -52.6
OFos-Solar +178.2	Gas-Wind -49.9
Bioenergy-Hydro +172.9	Coal-Gas +47.8
Bioenergy-OFos -147.1	Coal-OFos -18.7
Bioenergy-Coal +117.7	Gas-OFos -17.1
Nuclear-Solar +117.2	Hydro-Solar +9.2
Bioenergy-Gas +115.5	Hydro-Nuclear -4.8
Bioenergy-Wind -111.8	Coal-Hydro +3.8
OFos-Wind -104.5	Gas-Hydro +3.4
Nuclear-OFos -90.9	Coal-Solar +0.6
Nuclear-Wind -90.2	Gas-Solar +0.6
Solar-Wind -89.5	Coal-Nuclear -0.3
Hydro-Wind -87.1	Gas-Nuclear -0.3

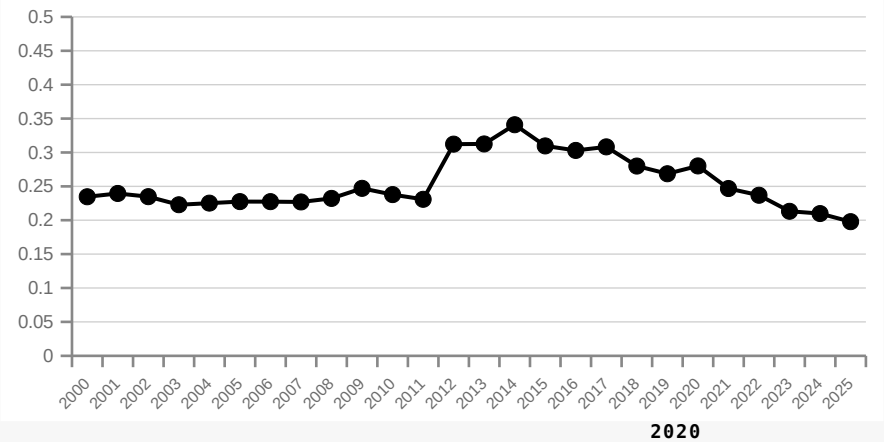
METRIC LEDGER

```
Hs     = 0.268630   Ring = Hs-2
E_metric = 12.030128   kappa = 55.89
omega   = 7.6236 deg
d_A     = 0.467986
Helm    = OFos delta = -0.321852

CLR:
 Bio -0.876 Coal +1.667 Gas +1.840 Hyd +0.110
 Nuc -0.009 OFos -0.566 Sol +0.018 Wind -2.184

Steering (1/x_j):
 Bio  37.9 Coal  3.0 Gas   2.5 Hyd  14.2
 Nuc  16.0 OFos 27.8 Sol  15.5 Wind 140.4
```

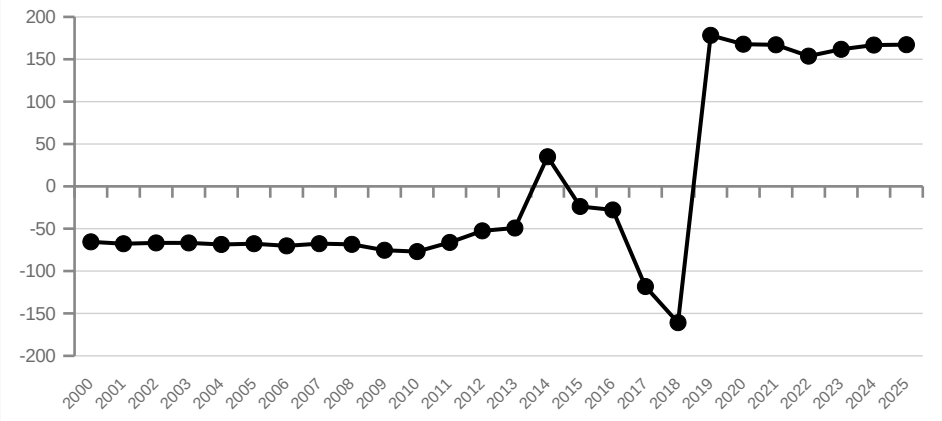
SECTION PLATE t=20 | Year 2020

Hs = 0.2801 | Hs-2 | omega = 9.79 deg

YZ: Hs vs Time



XZ: Bearing (OFos-Sola) vs Time



XY PLAN VIEW: All Bearings (deg)

OFos-Solar +167.8	Hydro-OFos -75.9
Bioenergy-Solar +164.8	Hydro-Nuclear -58.6
Bioenergy-Hydro +162.5	Coal-Wind -49.8
Bioenergy-Nuclear -152.6	Coal-Gas +48.0
Nuclear-Solar +152.3	Gas-Wind -46.8
Bioenergy-OFos -128.6	Hydro-Solar +40.8
OFos-Wind -114.2	Coal-OFos -28.0
Bioenergy-Coal +113.0	Gas-OFos -25.6
Nuclear-OFos -112.4	Coal-Nuclear -12.4
Bioenergy-Gas +111.0	Gas-Nuclear -11.2
Bioenergy-Wind -109.8	Coal-Hydro +7.6
Nuclear-Wind -100.5	Gas-Hydro +6.9
Solar-Wind -84.4	Coal-Solar +6.6
Hydro-Wind -83.5	Gas-Solar +5.9

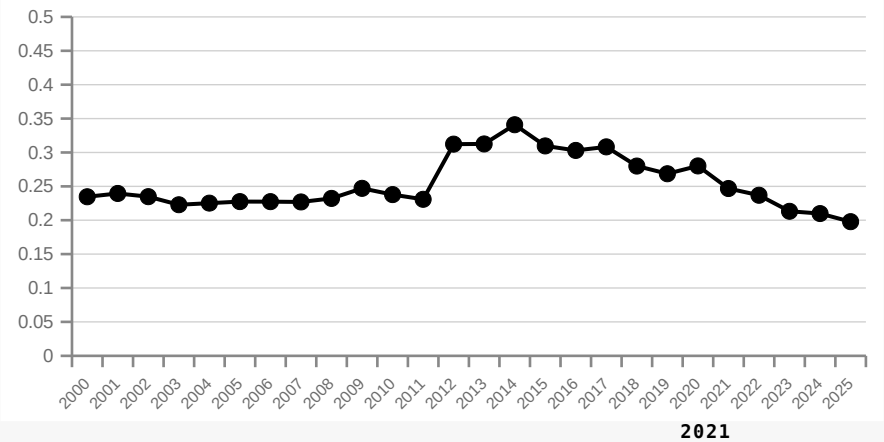
METRIC LEDGER

```
Hs     = 0.280146   Ring = Hs-2
E_metric = 11.994041   kappa = 48.68
omega   = 9.7926 deg
d_A     = 0.591660
Helm    = Nuc delta = -0.363464

CLR:
 Bio -0.720  Coal +1.696  Gas +1.881  Hyd +0.227
 Nuc -0.373  OFos -0.903  Sol +0.196  Wind -2.004

Steering (1/x_j):
 Bio   33.3 Coal   3.0 Gas   2.5 Hyd   12.9
 Nuc   23.5 OFos  39.9 Sol  13.3 Wind 120.2
```

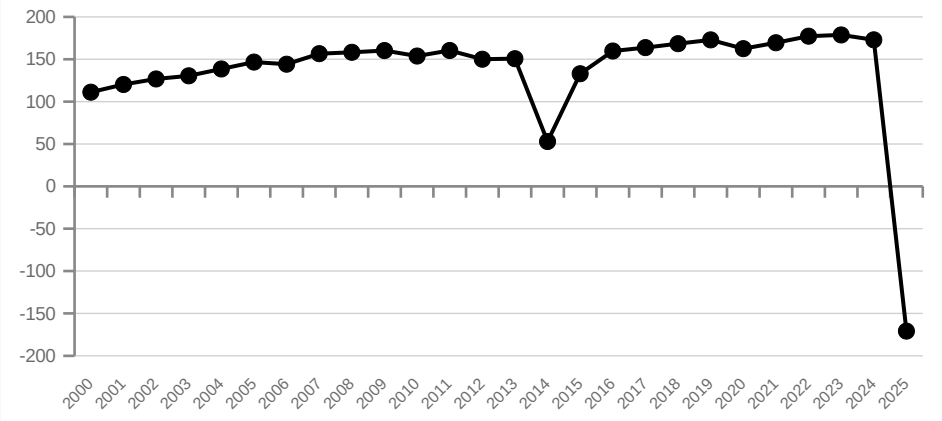
SECTION PLATE t=21 | Year 2021

Hs = 0.2468 | Hs-2 | omega = 4.71 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro	+169.5	Hydro-OFos	-80.8
Bioenergy-Nuclear	-169.2	Hydro-Solar	+55.0
OFos-Solar	+167.0	Coal-Wind	-51.1
Bioenergy-Solar	+165.1	Gas-Wind	-49.4
Bioenergy-OFos	-130.9	Coal-Gas	+46.8
Nuclear-Solar	+125.7	Hydro-Nuclear	-45.7
Bioenergy-Coal	+113.5	Coal-OFos	-26.6
Bioenergy-Gas	+112.2	Gas-OFos	-25.2
OFos-Wind	-112.0	Coal-Solar	+6.6
Bioenergy-Wind	-109.3	Gas-Solar	+6.2
Nuclear-OFos	-99.4	Coal-Nuclear	-4.7
Nuclear-Wind	-93.8	Coal-Hydro	+4.6
Hydro-Wind	-86.3	Gas-Nuclear	-4.5
Solar-Wind	-84.7	Gas-Hydro	+4.3

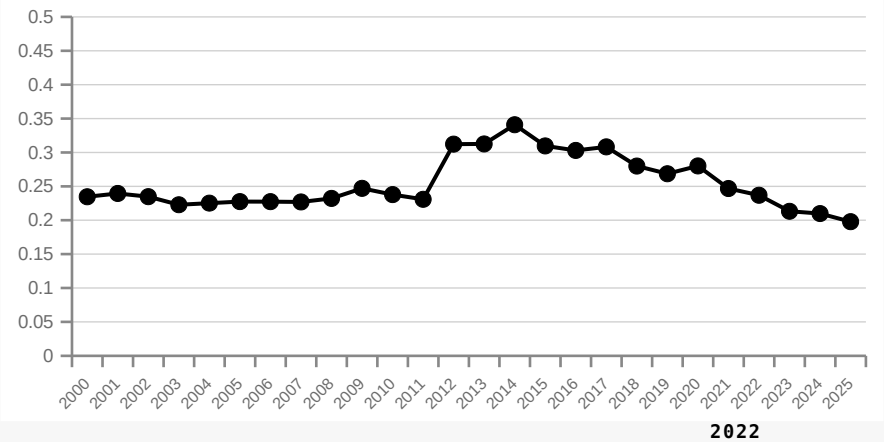
METRIC LEDGER

```
Hs     = 0.246834   Ring = Hs-2
E_metric = 10.666199   kappa = 40.44
omega   = 4.7105 deg
d_A     = 0.339626
Helm    = Nuc delta = +0.239503

CLR:
 Bio -0.698  Coal +1.605  Gas +1.708  Hyd +0.130
 Nuc -0.133  OFos -0.805  Sol +0.185  Wind -1.992

Steering (1/x_j):
 Bio   29.7 Coal   3.0 Gas   2.7 Hyd   13.0
 Nuc   16.9 OFos  33.1 Sol  12.3 Wind 108.5
```

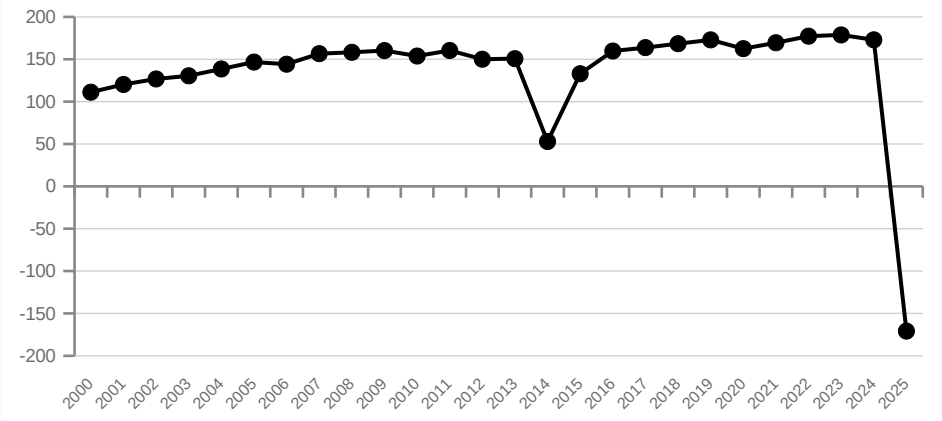
SECTION PLATE t=22 | Year 2022

Hs = 0.2369 | Hs-2 | omega = 7.43 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro +177.2	Hydro-Nuclear -84.6
Bioenergy-Solar +160.7	Solar-Wind -83.6
OFos-Solar +153.7	Hydro-Solar +82.1
Bioenergy-Nuclear -152.6	Coal-Wind -52.3
Nuclear-Solar +146.0	Gas-Wind -51.0
Bioenergy-OFos -144.7	Coal-Gas +46.4
Nuclear-OFos -126.2	Coal-OFos -16.3
Bioenergy-Coal +112.4	Gas-OFos -15.6
Bioenergy-Gas +111.5	Coal-Nuclear -12.1
Bioenergy-Wind -107.7	Gas-Nuclear -11.5
OFos-Wind -102.7	Coal-Solar +8.2
Nuclear-Wind -99.4	Gas-Solar +7.8
Hydro-Wind -89.1	Coal-Hydro +1.2
Hydro-OFos -86.0	Gas-Hydro +1.1

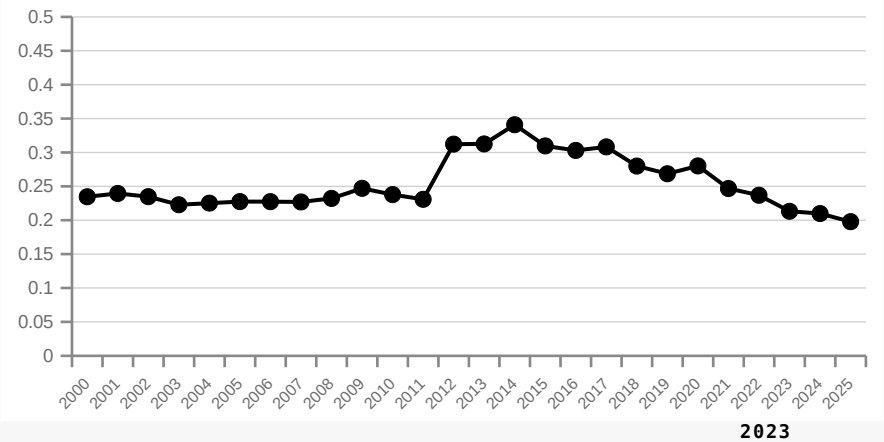
METRIC LEDGER

```
Hs     = 0.236919   Ring = Hs-2
E_metric = 10.187233   kappa = 40.22
omega   = 7.4261 deg
d_A     = 0.424692
Helm    = OFos delta = +0.344772

CLR:
 Bio -0.651  Coal +1.577  Gas +1.653  Hyd +0.032
 Nuc -0.337  OFos -0.460  Sol +0.228  Wind -2.041

Steering (1/x_j):
 Bio  27.5 Coal  3.0 Gas  2.7 Hyd  13.9
 Nuc  20.1 OFos  22.7 Sol  11.4 Wind 110.5
```

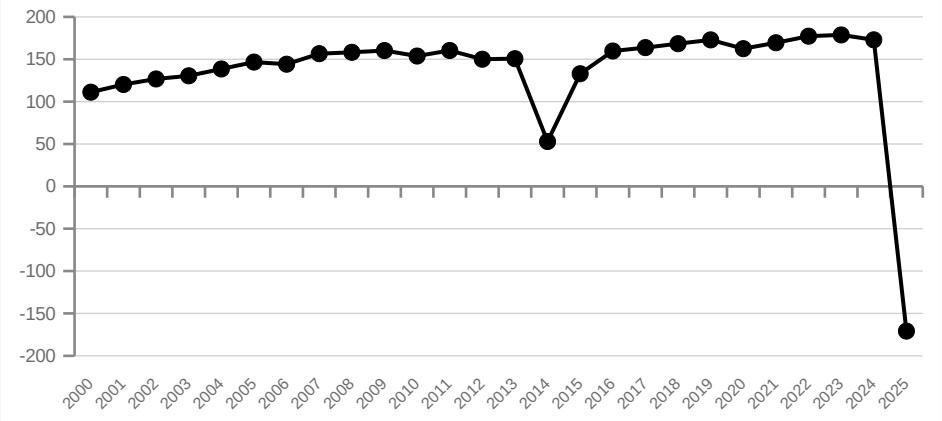
SECTION PLATE t=23 | Year 2023

Hs = 0.2132 | Hs-2 | omega = 10.00 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro +178.7	Solar-Wind -82.3
Bioenergy-Nuclear +174.7	Nuclear-Solar +79.2
OFos-Solar +161.7	Hydro-Nuclear +75.9
Bioenergy-Solar +154.3	Coal-Wind -53.3
Bioenergy-OFos -124.4	Gas-Wind -52.2
OFos-Wind -112.2	Coal-Gas +46.1
Bioenergy-Coal +110.6	Coal-OFos -28.7
Bioenergy-Gas +109.9	Gas-OFos -27.8
Bioenergy-Wind -105.7	Coal-Solar +10.3
Hydro-Wind -89.6	Gas-Solar +9.9
Hydro-OFos -89.1	Coal-Nuclear +2.0
Nuclear-Wind -88.5	Gas-Nuclear +1.9
Hydro-Solar +87.2	Coal-Hydro +0.5
Nuclear-OFos -86.4	Gas-Hydro +0.5

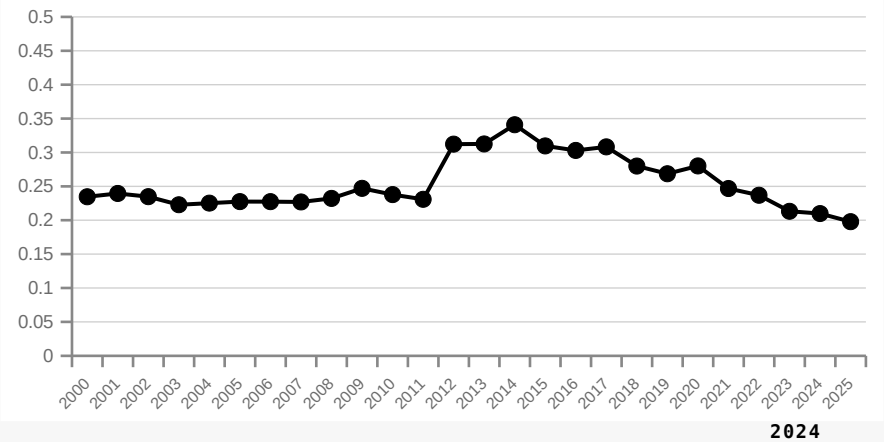
METRIC LEDGER

```
Hs    = 0.213214   Ring = Hs-2
E_metric = 9.686031   kappa = 34.84
omega   = 10.0007 deg
d_A    = 0.555145
Helm    = Nuc delta = +0.388708

CLR:
 Bio -0.560 Coal +1.492 Gas +1.551 Hyd +0.013
 Nuc +0.052 OFos -0.817 Sol +0.270 Wind -2.000

Steering (1/x_j):
 Bio   24.0 Coal   3.1 Gas   2.9 Hyd   13.5
 Nuc   13.0 OFos  31.0 Sol  10.4 Wind 101.1
```

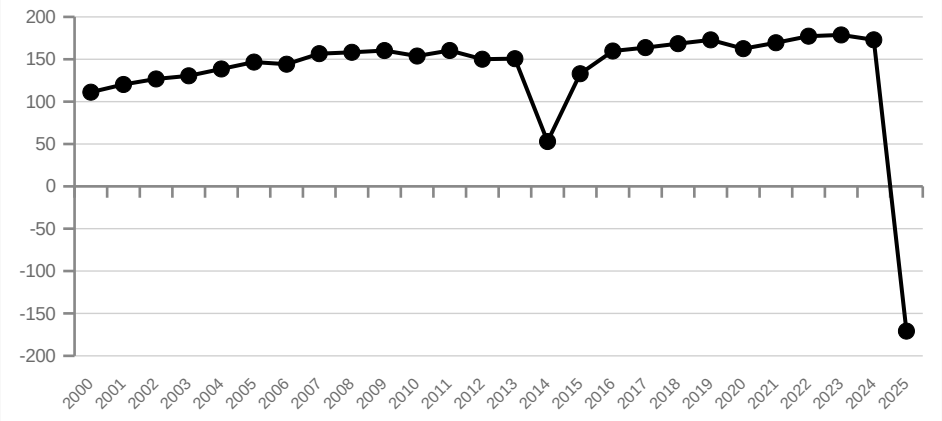
SECTION PLATE t=24 | Year 2024

Hs = 0.2099 | Hs-2 | omega = 6.04 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro	+172.9	Hydro-Solar	+76.5
OFos-Solar	+166.7	Hydro-Nuclear	+64.3
Bioenergy-Nuclear	+165.5	Nuclear-Solar	+63.6
Bioenergy-Solar	+152.6	Coal-Wind	-51.7
OFos-Wind	-120.4	Gas-Wind	-50.6
Bioenergy-OFos	-114.4	Coal-Gas	+46.1
Bioenergy-Coal	+108.6	Coal-OFos	-36.5
Bioenergy-Gas	+108.0	Gas-OFos	-35.5
Bioenergy-Wind	-104.9	Coal-Solar	+9.9
Hydro-Wind	-88.1	Gas-Solar	+9.6
Hydro-OFos	-86.8	Coal-Nuclear	+5.0
Nuclear-Wind	-86.1	Gas-Nuclear	+4.8
Nuclear-OFos	-83.3	Coal-Hydro	+2.4
Solar-Wind	-82.1	Gas-Hydro	+2.3

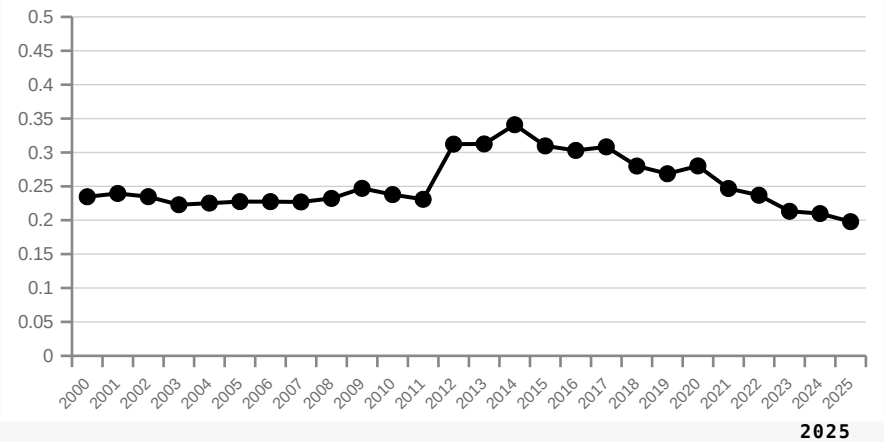
METRIC LEDGER

```
Hs     = 0.209872   Ring = Hs-2
E_metric = 9.553951   kappa = 30.00
omega   = 6.0418 deg
d_A     = 0.327601
Helm    = OFos delta = -0.277449

CLR:
 Bio -0.497  Coal +1.477  Gas +1.534  Hyd +0.062
 Nuc +0.128  OFos -1.094  Sol +0.258  Wind -1.867

Steering (1/x_j):
 Bio   22.4 Coal   3.1 Gas   2.9 Hyd   12.8
 Nuc   12.0 OFos  40.7 Sol  10.5 Wind  88.1
```

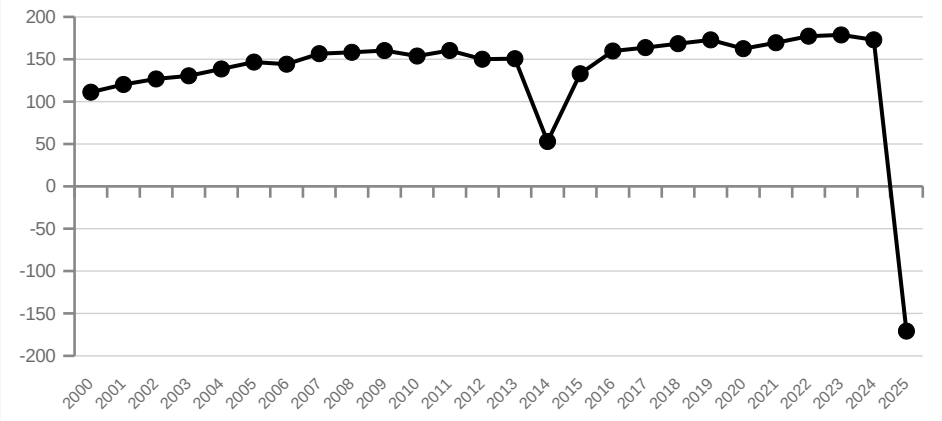
SECTION PLATE t=25 | Year 2025

Hs = 0.1978 | Hs-2 | omega = 3.76 deg

YZ: Hs vs Time



XZ: Bearing (Bioe-Hydr) vs Time



XY PLAN VIEW: All Bearings (deg)

Bioenergy-Hydro -170.9	Solar-Wind -81.9
OFos-Solar +167.2	Nuclear-OFos -80.7
Bioenergy-Nuclear +152.4	Nuclear-Solar +54.2
Bioenergy-Solar +144.1	Coal-Wind -51.4
OFos-Wind -121.9	Gas-Wind -51.0
Bioenergy-OFos -107.5	Coal-Gas +45.4
Hydro-Nuclear +107.1	Coal-OFos -38.0
Bioenergy-Coal +103.8	Gas-OFos -37.6
Bioenergy-Gas +103.6	Coal-Solar +10.1
Hydro-Solar +102.5	Gas-Solar +9.9
Bioenergy-Wind -101.1	Coal-Nuclear +7.3
Hydro-OFos -92.9	Gas-Nuclear +7.2
Hydro-Wind -91.8	Coal-Hydro -2.3
Nuclear-Wind -84.2	Gas-Hydro -2.2

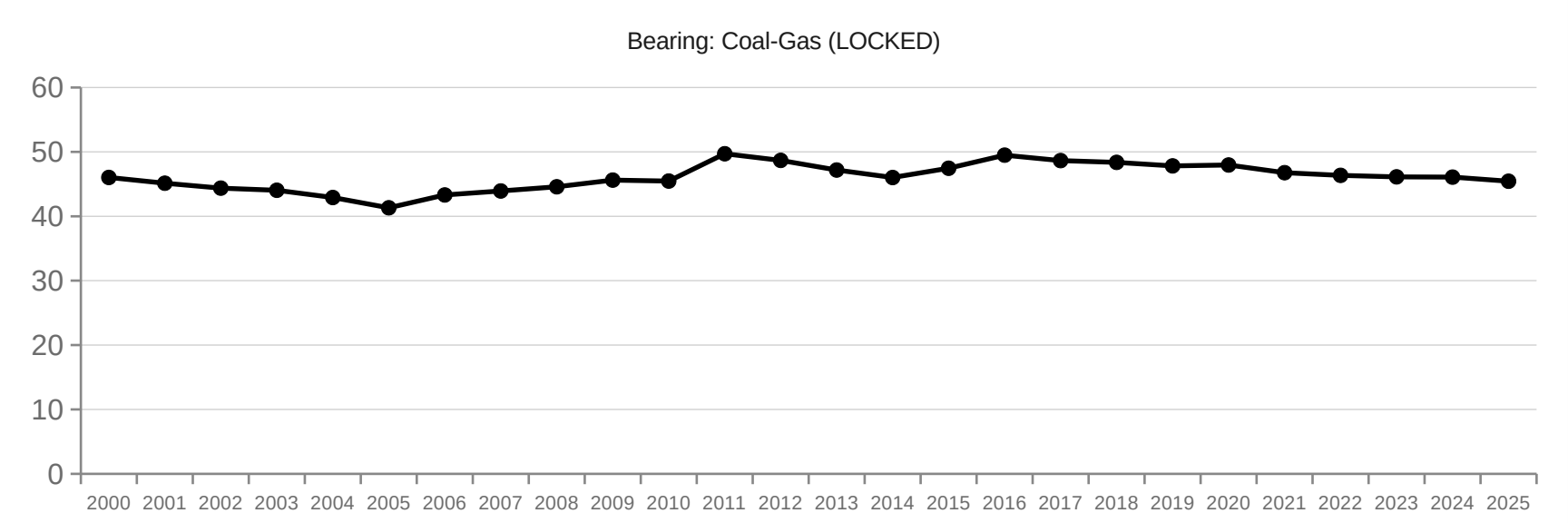
METRIC LEDGER

```
Hs    = 0.197825   Ring = Hs-2
E_metric = 8.967570   kappa = 26.27
omega   = 3.7598 deg
d_A    = 0.221651
Helm    = Bio delta = +0.143563

CLR:
 Bio -0.354  Coal +1.440  Gas +1.462  Hyd -0.057
 Nuc +0.185  OFos -1.125  Sol +0.256  Wind -1.806

Steering (1/x_j):
 Bio   18.8 Coal   3.1 Gas   3.1 Hyd   13.9
 Nuc   10.9 OFos  40.6 Sol  10.2 Wind  80.1
```

INFORMATIONAL LOCK DETECTION

Carrier pairs with bearing spread < 10 deg across all years

LOCKED: Coal-Gas spread = 8.4 deg



Section Cine-Deck Complete

26 section plates (1 per year)

28 bearing pairs per plate

Locks detected: 1

Source: stage1_output.json

The instrument reads. The expert decides. The loop stays open.